

22050 Continuous-Time Signals and Linear Systems

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L12

Butterworth filter design – part 1

Defining properties

Pole symmetries in Butterworth filters

Sallen-Key circuit as a Butterworth filter

Sensitivity analysis

Sensitivity-optimized filter design

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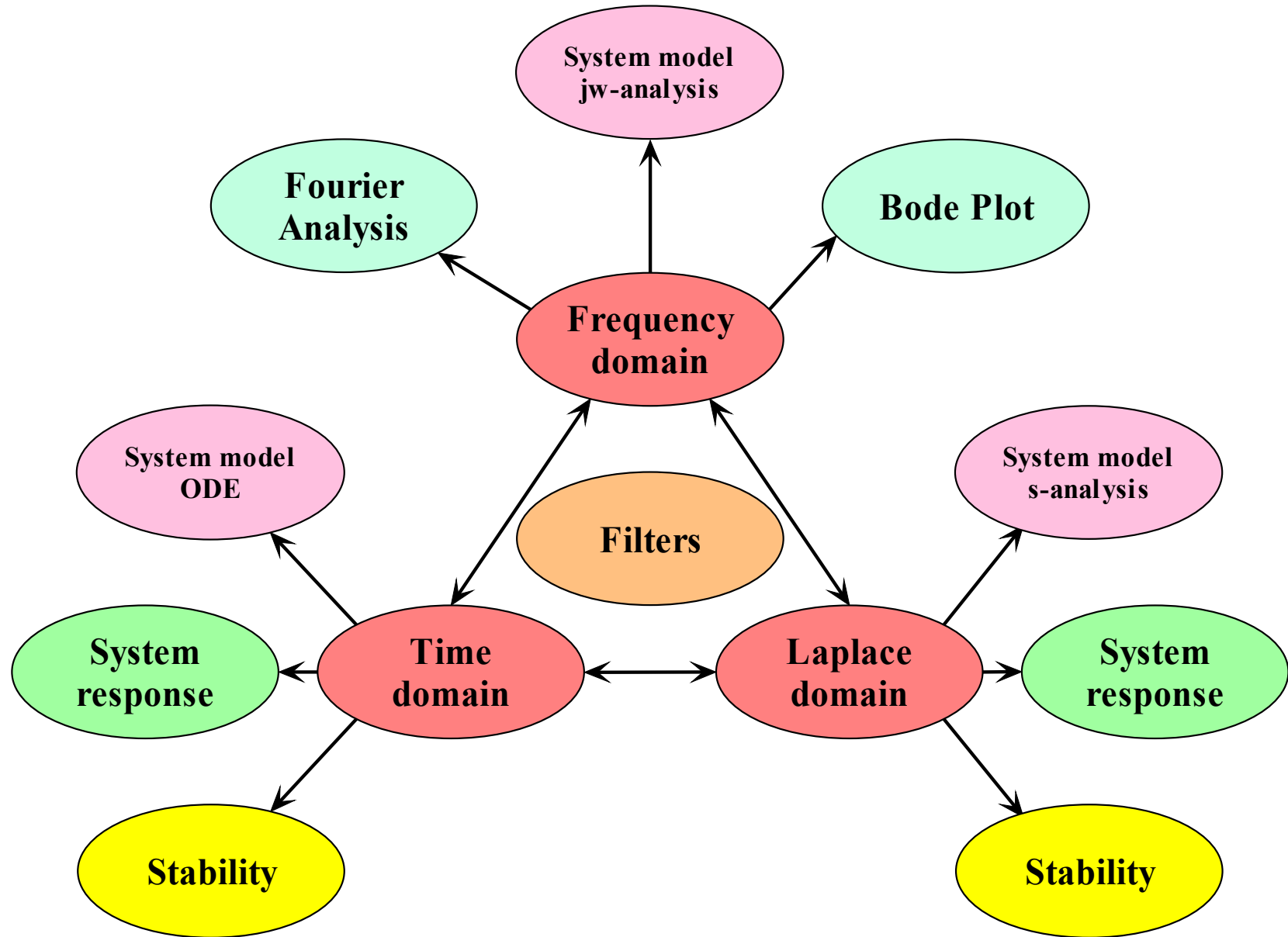
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- Defining properties of Butterworth filters
- Butterworth Transfer function (Lathi 7.5)
 - Defining equation
 - Pole positions and symmetries
 - Calculating coefficients
- Design of Butterworth lowpass filter using a Sallen-Key circuit
- Sensitivity to change in component values
- Strategy for sensitivity-optimized filter design
- Designing a sensitivity-optimized frequency-normalized lowpass filter
- Frequency scaling
- Impedance scaling

Butterworth filters

Defining properties

Video

The objective of a filter is to let frequencies in the passband through the filter with a specified maximum variation in gain (from G_0 to G_p), while attenuating signals in the stopband by a specified attenuation G_s .

G_0 : Gain at $\omega = 0$

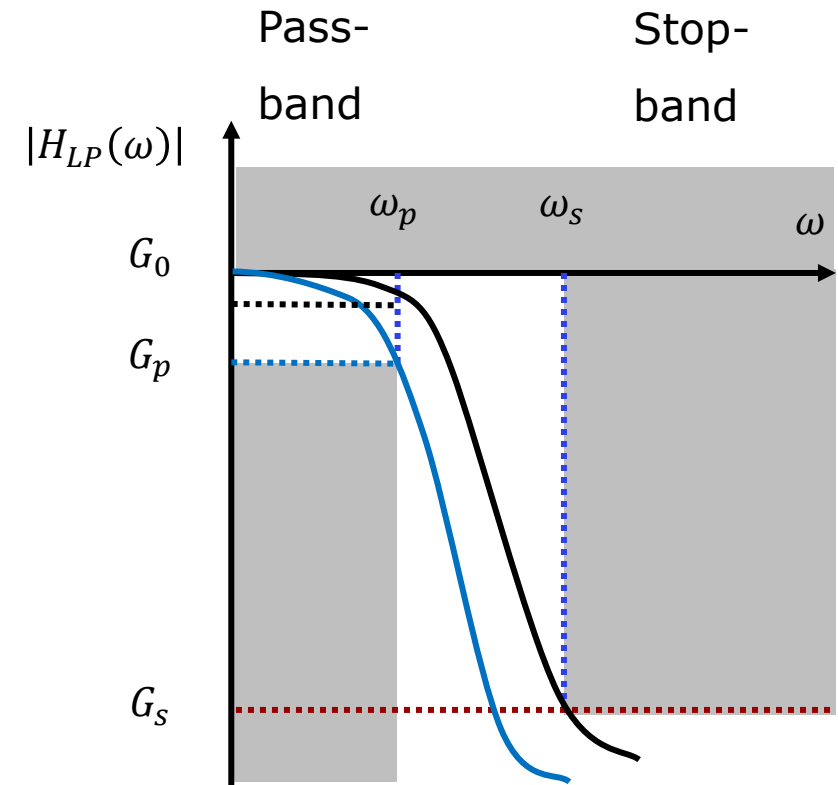
G_p : Gain at $\omega = \omega_p$

G_s : Gain at $\omega = \omega_s$

The gain at DC can be set as wanted, but most often it is set to 1 (0dB).

The gain at the end of the passband can be set as wanted. If it is set to -3dB, then $\omega_p = \omega_c = \omega_{3dB}$.

ω_p does not have to be the 3dB cut-off frequency. It could fx be the frequency, where the gain has decreased by 1dB. In this case $\omega_c > \omega_p$.



The blue curve satisfies specifications for the attenuation G_p at ω_p .

The black curve satisfies specifications for the attenuation G_s at ω_s .

The amplitude response of an n'th order **Butterworth filter** is by definition:

The gain at the cut-off frequency ω_c is – 3dB regardless of the filter order n.

We notice that the defining equation produces a **lowpass** frequency response. Other types of filters (highpass, bandpass, stopband) can be obtained by frequency transformation methods. Even in these cases we start by designing a lowpass filter.

It appear at first glance that the above definition ignores completely the phase characteristic. We will discover, that this is not the case.

$$|H(j\omega)| \stackrel{\text{def}}{=} \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

$$|H(j\omega_c)| \stackrel{\text{def}}{=} \frac{1}{\sqrt{1 + (1)^{2n}}} = \frac{1}{\sqrt{2}}$$

Defining properties of the Butterworth filter

To be general, we add a pass-band gain parameter ϵ . We also substitute $\omega_p \rightarrow \omega_c$.

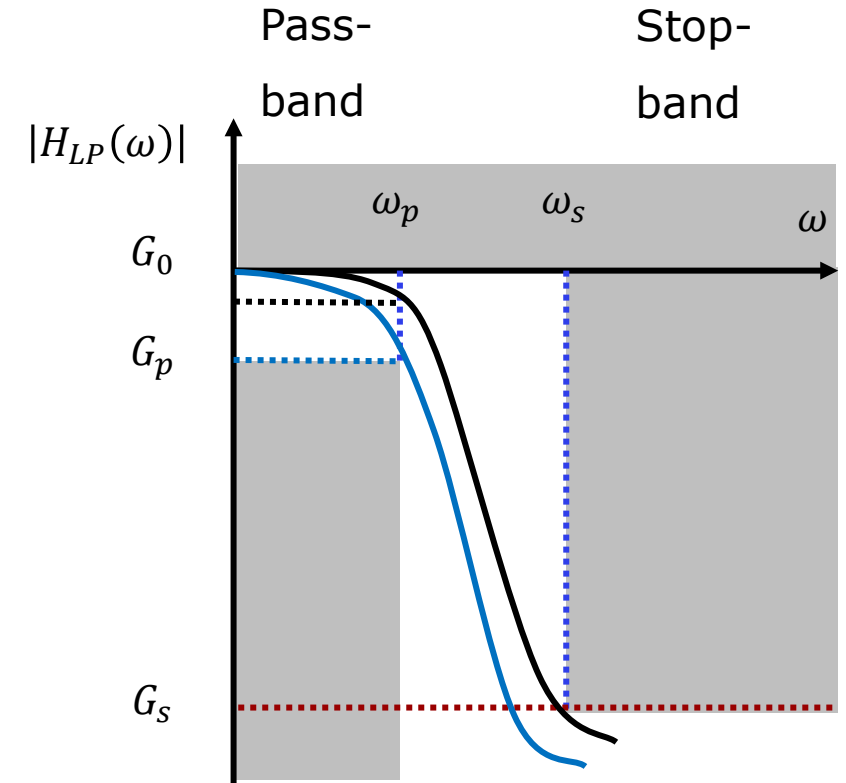
We define the magnitude squared:

This function can be expanded into a Maclaurin series, which is a Taylor series expanded about $\omega = 0$.

$$|H(j\omega)| \stackrel{\text{def}}{=} \frac{1}{\sqrt{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2n}}}$$

$$G(j\omega) \stackrel{\text{def}}{=} \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^{2n}}$$

$$G(j\omega) = \sum_{k=0}^{\infty} (-1)^k \epsilon^{2k} \left(\frac{\omega}{\omega_p}\right)^{2nk}$$



For a first order filter:

$$G(j\omega) = \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^2} = 1 - \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^2 + \epsilon^4 \left(\frac{\omega}{\omega_p}\right)^4 - \epsilon^6 \left(\frac{\omega}{\omega_p}\right)^6 + \dots$$

For a second order filter:

$$G(j\omega) = \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^4} = 1 - \epsilon^2 \left(\frac{\omega}{\omega_p}\right)^4 + \epsilon^4 \left(\frac{\omega}{\omega_p}\right)^8 - \epsilon^6 \left(\frac{\omega}{\omega_p}\right)^{12} + \dots$$

We want to investigate the **slope** of the squared magnitude function at DC.

For convenience, all coefficients are collected in A_k :

Taking the m 'th derivative:

If $m < 2n$ then all terms will contain ω to a non-zero power.

If $m = 2n$:

We observe that the Butterworth squared magnitude function has zero slope at DC to the $2n-1$ derivative.

$$G(j\omega) = \sum_{k=0}^{\infty} (-1)^k \epsilon^{2k} \left(\frac{\omega}{\omega_p} \right)^{2nk} = \sum_{k=0}^{\infty} A_k \omega^{2nk}$$

$$G(j\omega) = A_0 + A_1 \omega^{2n} + A_2 \omega^{4n} + \dots$$

$$G^{(m)}(j\omega) = B_1 \omega^{2n-m} + B_2 \omega^{4n-m} + \dots$$

$$G^{(m)}(j0) = 0, \quad m < 2n$$

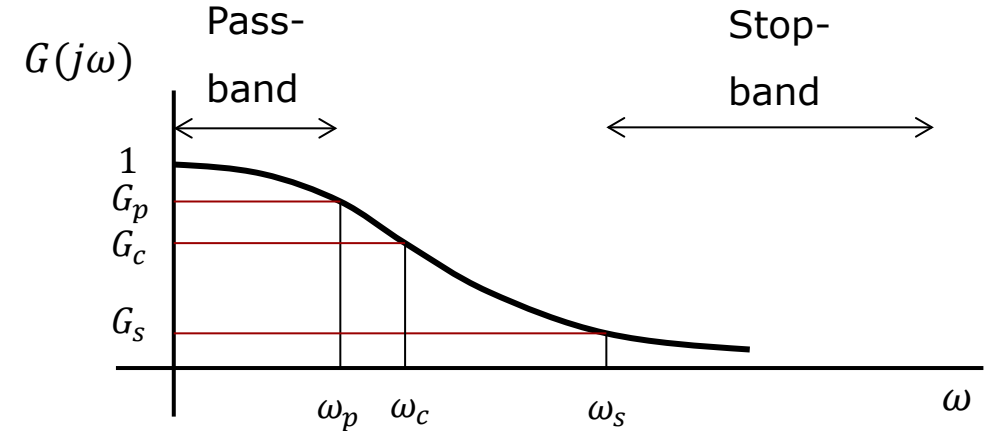
$$G^{(m=2n)}(j\omega) = B_1 \omega^0 + B_2 \omega^{4n-2n} + \dots$$

$$G^{(m)}(j0) = B_1, \quad m = 2n$$

The Butterworth filter has a maximally flat amplitude response at $\omega = 0$.

This is the Hallmark of the Butterworth filter.

We now apply the passband and stopband specifications to the squared magnitude.



$$G(j\omega) \stackrel{\text{def}}{=} \frac{1}{1 + \epsilon^2 \left(\frac{\omega}{\omega_p} \right)^{2n}}$$

$$\frac{1}{1 + \epsilon^2} < G(j\omega) < 1, \quad 0 < \omega < \omega_p$$

$$G(j\omega) < \frac{1}{1 + \epsilon^2 \left(\frac{\omega_s}{\omega_p} \right)^{2n}}, \quad \omega > \omega_s$$

The specifications for the **passband** states how much gain is allowed to vary within a specified passband.

The specifications for the **stopband** states the minimum attenuation for signal frequencies within the stopband.

Using the gain parameter ϵ we can specify, how much the gain is allowed to vary in the frequency range $0 < \omega < \omega_p$, independent of filter order n .

The frequency ω_p is that frequency, where the squared magnitude is:

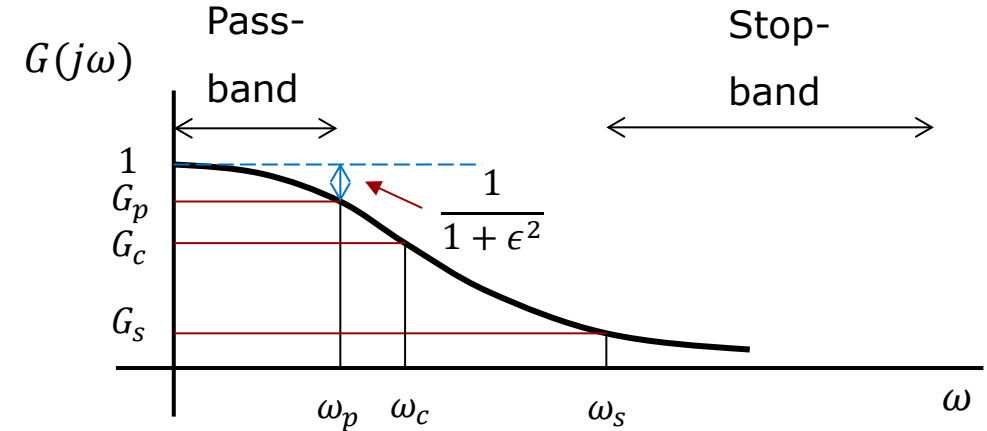
$\epsilon^2 = 0.1$: Squared magnitude drops from 1 to 0.9

$\epsilon^2 = 1$: Squared magnitude drops from 1 to 0.5

Magnitude drops from 1 to $\frac{1}{\sqrt{2}}$, from 0dB to -3dB

If we want ω_p to be the 3dB cut-off frequency (ω_c), then we must set $\epsilon = 1$ and the squared magnitude $G_p = 1/2$.

This is the most used choice for the Butterworth filter. This is equivalent to defining:



$$G(j\omega_p) \stackrel{\text{def}}{=} \frac{1}{1 + \epsilon^2 \left(\frac{\omega_p}{\omega_p}\right)^{2n}} = \frac{1}{1 + \epsilon^2}$$

$$G(j\omega) \stackrel{\text{def}}{=} \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}$$

$$\epsilon \stackrel{\text{def}}{=} \left(\frac{\omega_p}{\omega_c}\right)^n$$

Butterworth filters

Transfer function

Video

The design procedure uses tables of filter transfer function coefficients. To minimize the need for such tables, all filters, regardless of cut-off frequency, is initially designed as a **frequency-normalized lowpass filter with a cut-off frequency of 1 rad/s**.

For this reason, we only need one coefficient table for all types of Butterworth filters.

We cannot design a system from the amplitude response function ($|H(j\omega)|$) alone because that would make the phase arbitrary.

We need to consider the **squared magnitude** $G(j\omega)$ of the defining function.

We have set $\epsilon = 1$ so that we can define $\omega_p = \omega_c$ as the 3dB cut-off frequency.

For further insight we need to study the poles of $G(s)$.

$$G(j\omega) \stackrel{\text{def}}{=} \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}$$

$$G(j\omega) \stackrel{\text{def}}{=} \frac{1}{1 + \left(\frac{j\omega}{j\omega_c}\right)^{2n}}$$

$$G(s) = \frac{1}{1 + \left(\frac{s}{j\omega_c}\right)^{2n}}$$

We will seek the transfer function for the frequency normalized lowpass filter, i.e., $\omega_c \stackrel{\text{def}}{=} 1\text{rad/s}$

$$G(s) = \frac{1}{1 + \left(\frac{s}{j\omega_c}\right)^{2n}}$$

$$G(s) = \frac{1}{1 + \left(\frac{s}{j}\right)^{2n}}$$

The denominator has poles when :

$$\begin{aligned} \left(\frac{s}{j}\right)^{2n} &= -1 \Rightarrow s_k^{2n} = -(j)^{2n} \\ &= e^{j\pi(2k-1)} \times \left(e^{\frac{j\pi}{2}}\right)^{2n} \\ &= e^{j\pi(2k+n-1)} \end{aligned}$$

Taking the $2n$ root on both sides, we get the **poles**:

$G(s)$ has $2n$ poles lying evenly distributed on the **unit circle**.

$$s_k = \underset{\substack{\uparrow \\ \text{radius}}}{1} \cdot e^{j\frac{\pi}{2n}(2k+n-1)}, \quad k = 1, 2, 3, \dots, 2n$$

↑
angle

$$G(s) = \frac{1}{1 + \left(\frac{s}{j}\right)^{2n}}$$

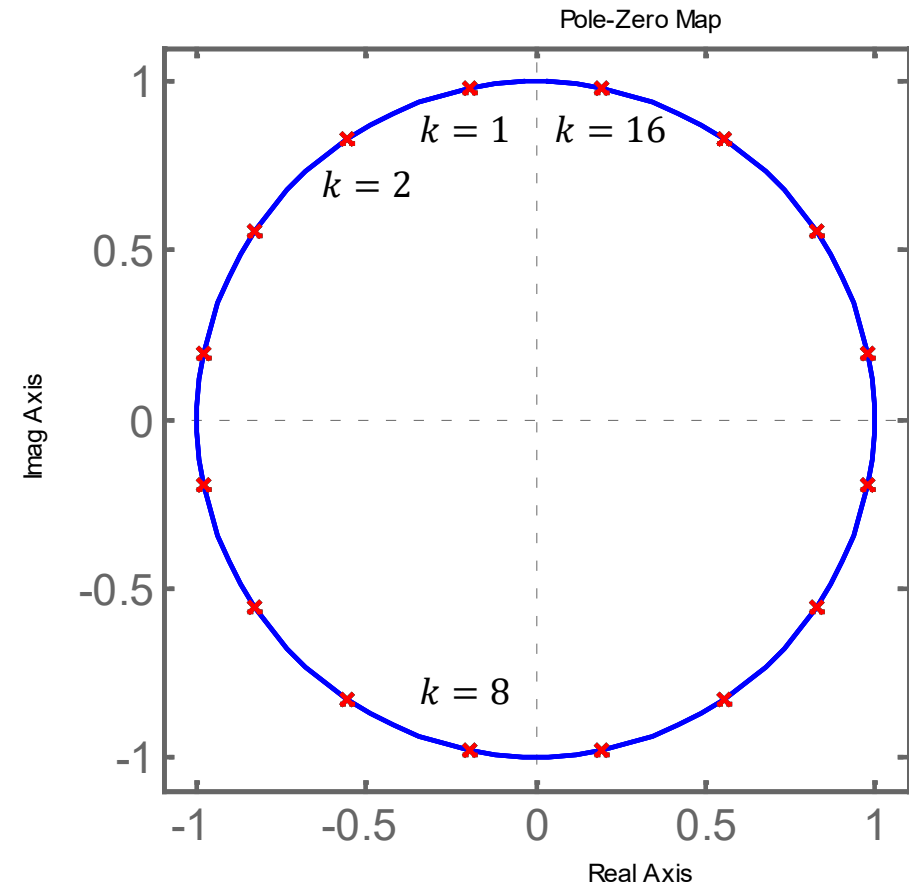
$$s_k = 1 \cdot e^{j\frac{\pi}{2n}(2k+n-1)}, \quad k = 1, 2, 3, \dots, 2n$$

Observation:

The poles of $G(s)$ are evenly distributed on the unit circle.

In the interest of stability, we cannot implement a filter transfer function $H(s)$ that contains **poles in the right half-plane**.

We need to **split up the poles in two groups**: those that lie in the left half-plane and those that lie in the right half-plane.



Pole symmetries with only 4 poles

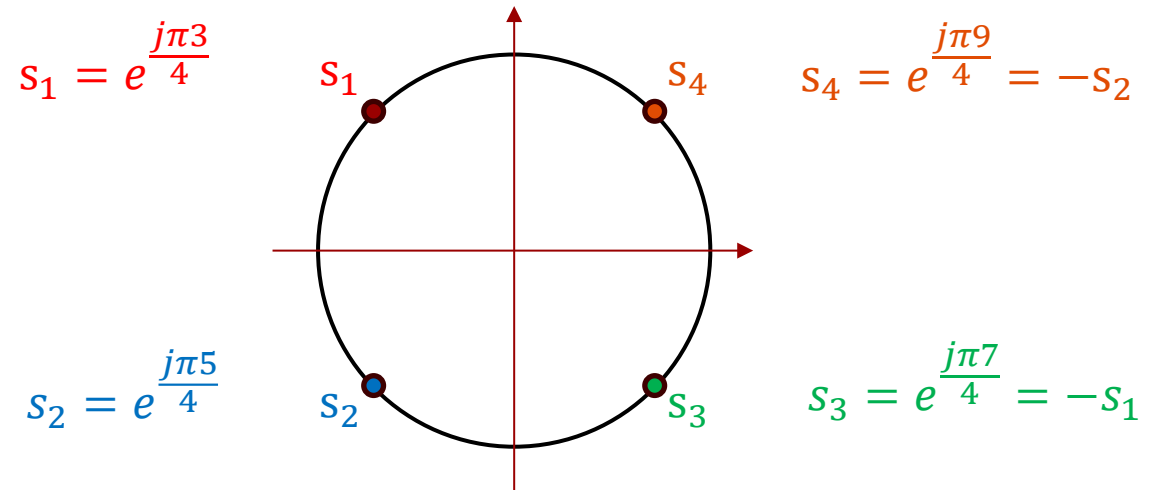
With 4 poles we see the following symmetries.

We observe that s_1 and s_2 are complex conjugated poles in the left half-plane.

We observe that s_3 and s_4 are complex conjugated poles in the right half-plane.

We also see that each pole in the right half-plane is the negative of a **diagonal twin** in the left half-plane.

We recognize $H(s)$ as the Laplace transform of $h(t)$. But how do we account for $H(-s)$?



$$G(s) = \underbrace{\frac{1}{(s - s_1)(s - s_2)}}_{\text{left half-plane}} \underbrace{\frac{1}{(s - s_3)(s - s_4)}}_{\text{right half-plane}}$$

$$G(s) = \underbrace{\frac{1}{(s - s_1)(s - s_2)}}_{\text{left half-plane}} \underbrace{\frac{1}{(s + s_1)(s + s_2)}}_{\text{right half-plane}}$$

$$G(s) = \underbrace{\frac{1}{(s - s_1)(s - s_2)}}_{\text{left half-plane}} \underbrace{\frac{1}{(-s - s_1)(-s - s_2)}}_{\text{right half-plane}} = \underbrace{H(s)}_{\text{left}} \underbrace{H(-s)}_{\text{right}}$$

The poles of $H(s)$ lies in the left half plane. The system is stable and has the **causal** impulse response:

Using the **bilateral Laplace transform**, we can find the Laplace transform of the time-reflected signal $h(-t)$:

We make the substitution $\tau = -t$ and $d\tau = -dt$:

Since reversing the integration limits introduces a negative sign, we get:

This result is a proof of the **time reversal theorem** for the bilateral Laplace transform.

It shows that the system with poles in the right half plane has an **anti-causal** impulse response.

$$h(t) \leftrightarrow H(s)$$

$$\mathcal{L}\{h(-t)\} = \int_{-\infty}^{\infty} h(-t)e^{-st} dt$$

$$\mathcal{L}\{h(-t)\} = \int_{\infty}^{-\infty} h(\tau)e^{s\tau}(-d\tau)$$

$$\mathcal{L}\{h(-t)\} = \int_{-\infty}^{\infty} h(\tau)e^{s\tau} d\tau$$

$$\mathcal{L}\{h(-t)\} = \int_{-\infty}^{\infty} h(\tau)e^{-(-s)\tau} d\tau = H(-s)$$

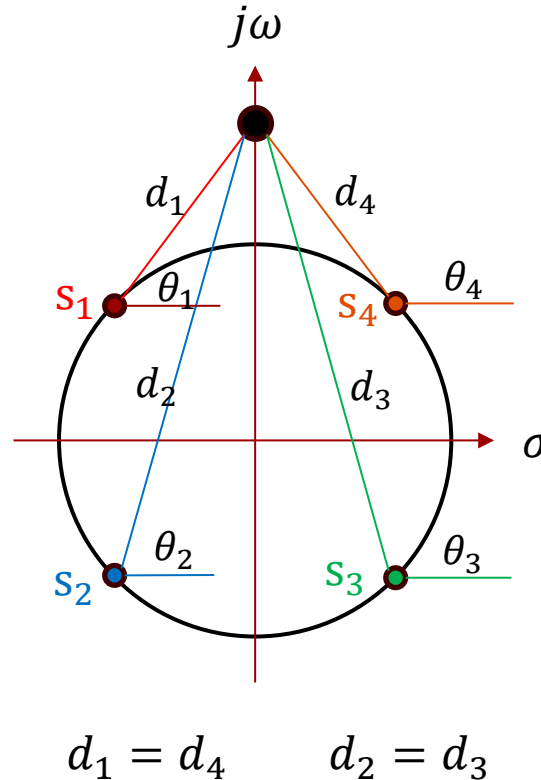
$$h(-t) \leftrightarrow H(-s)$$

The defining function $G(j\omega)$ had no phase.

We are now in a position to explain why.

From the sketch we observe that poles mirrored about the imaginary axis have phase angles that add up to π (180°).

It follows from the math that $H(j\omega)$ and $H(-j\omega)$ have phase angles that are different by π (180°). In other words, their angles add to zero.



$$H(j\omega) = \frac{1}{d_1 e^{j\theta_1}} \cdot \frac{1}{d_2 e^{j\theta_2}} = \frac{1}{d_1 d_2} e^{-j(\theta_1 + \theta_2)}$$

$$H(-j\omega) = \frac{1}{d_3 e^{j\theta_3}} \cdot \frac{1}{d_4 e^{j\theta_4}}$$

$$\theta_3 = \pi - \theta_2 \quad \theta_4 = \pi - \theta_1$$

$$\begin{aligned} H(-j\omega) &= \frac{1}{d_3 e^{j\theta_3}} \cdot \frac{1}{d_4 e^{j\theta_4}} \\ &= \frac{1}{d_2 e^{j(\pi - \theta_2)} d_1 e^{j(\pi - \theta_1)}} \\ &= \frac{1}{d_1 d_2} e^{j(\theta_1 + \theta_2)} \end{aligned}$$

$$G(j\omega) = H(j\omega)H(-j\omega) = \frac{1}{d_1 d_2} e^{-j(\theta_1 + \theta_2)} \frac{1}{d_1 d_2} e^{j(\theta_1 + \theta_2)} = \frac{1}{(d_1 d_2)^2}$$

Observation:

The function $G(j\omega)$ defining the Butterworth filter had zero phase.

We now know, that this does not imply that the transfer function for the Butterworth filter has zero phase, or that the phase is arbitrary.

It means that the **anti-causal twin** filter has a phase that cancels out the phase of the causal Butterworth filter.

The causal filter that we want to implement has a transfer function of the form:

$$G(j\omega) \stackrel{\text{def}}{=} \frac{1}{1 + \left(\frac{j\omega}{j\omega_c}\right)^{2n}}$$

$$\begin{aligned} G(j\omega) &= H(j\omega)H(-j\omega) \\ &= \frac{1}{d_1 d_2} e^{-j(\theta_1 + \theta_2)} \frac{1}{d_1 d_2} e^{j(\theta_1 + \theta_2)} \\ &= \frac{1}{(d_1 d_2)^2} \end{aligned}$$

$$H(j\omega) = \frac{1}{d_1 d_2} e^{-j(\theta_1 + \theta_2)}$$

$$H(s) = \frac{1}{(s - s_1)(s - s_2)}$$

Using only the poles in the left half-plane, k must run from 1 to n , not from 1 to $2n$:

The transfer function is:

$$s_k = e^{\frac{j\pi}{2n}(2k+n-1)}, \quad k = 1, 2, 3, \dots, n$$

$$H(s) = \frac{1}{(s - s_1)(s - s_2) \cdots (s - s_n)}$$

Pole indices

Order	n	k=1	k=2	k=3	k=4	k=5
	1	$e^{j\pi}$				
	2	$e^{j\pi 3/4}$	$e^{j\pi 5/4}$			
	3	$e^{j\pi 4/6}$	$e^{j\pi}$	$e^{j\pi 8/6}$		
	4	$e^{j\pi 5/8}$	$e^{j\pi 7/8}$	$e^{j\pi 9/8}$	$e^{j\pi 11/8}$	
	5	$e^{j\pi 6/10}$	$e^{j\pi 8/10}$	$e^{j\pi}$	$e^{j\pi 12/10}$	$e^{j\pi 14/10}$

Poles of Butterworth filter

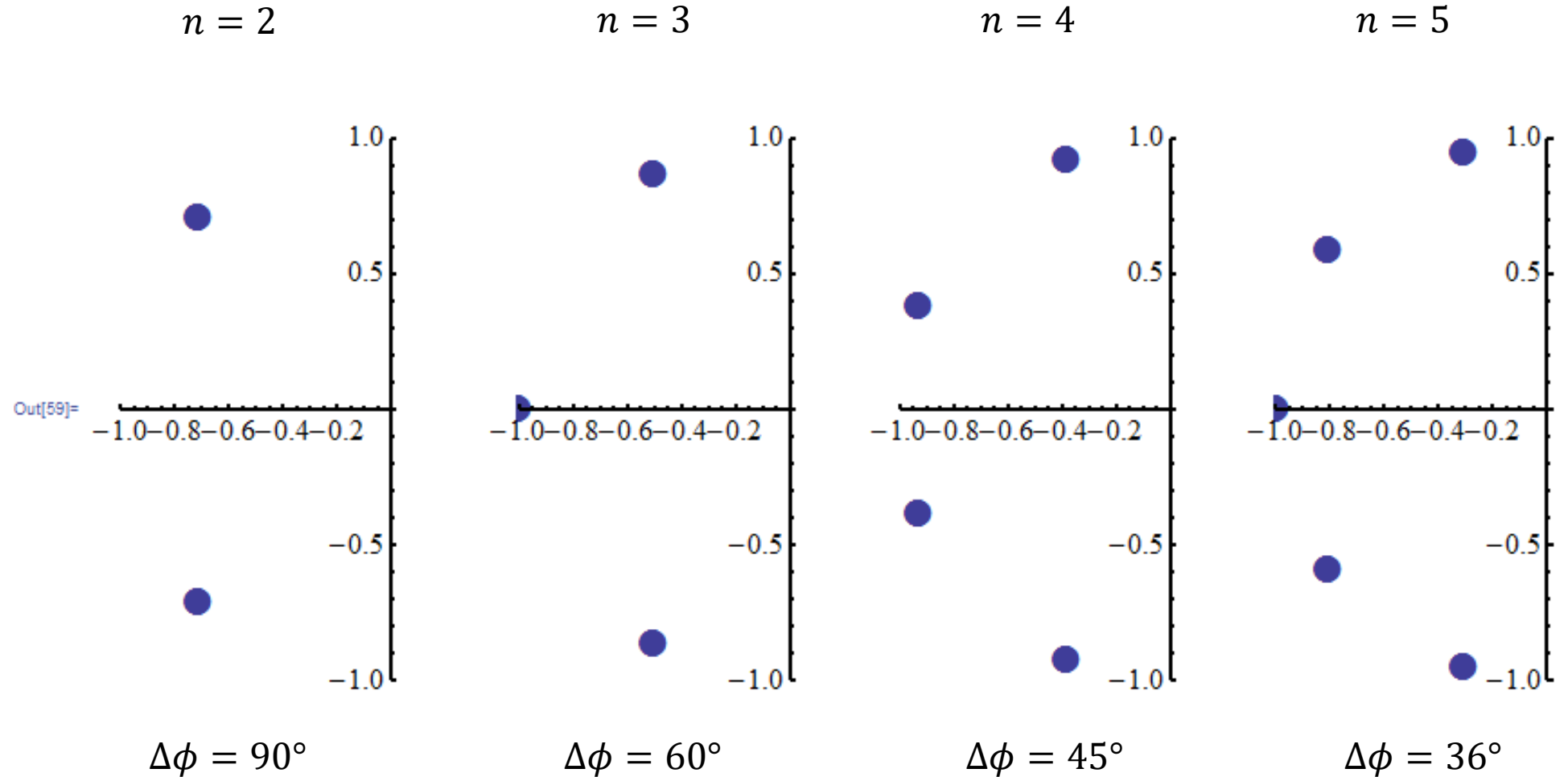
n	k=1	k=2	k=3	k=4	k=5
1	$e^{j\pi}$				
2	$e^{j\pi 3/4}$	$e^{j\pi 5/4}$			
3	$e^{j\pi 4/6}$	$e^{j\pi}$	$e^{j\pi 8/6}$		
4	$e^{j\pi 5/8}$	$e^{j\pi 7/8}$	$e^{j\pi 9/8}$	$e^{j\pi 11/8}$	
5	$e^{j\pi 6/10}$	$e^{j\pi 8/10}$	$e^{j\pi}$	$e^{j\pi 12/10}$	$e^{j\pi 14/10}$

Making use of pole symmetries:

n	k=1	k=2	k=3	k=4	k=5
1	$e^{j\pi}$				
2	$e^{j\pi 3/4}$	$e^{-j\pi 3/4}$			
3	$e^{j\pi 4/6}$	$e^{j\pi}$	$e^{-j\pi 4/6}$		
4	$e^{j\pi 5/8}$	$e^{j\pi 7/8}$	$e^{-j\pi 7/8}$	$e^{-j\pi 5/8}$	
5	$e^{j\pi 6/10}$	$e^{j\pi 8/10}$	$e^{j\pi}$	$e^{-j\pi 8/10}$	$e^{-j\pi 6/10}$

The angle between poles depends on n :

$$\Delta\phi = \frac{180}{n}$$



As many poles are complex conjugated, we can pair these factors in **quadratic factors**:

Linear factor: 1st order filter circuit.

Quadratic factor: 2nd order filter circuit, such as the **Multiple feedback circuit** or the **Sallen-Key circuit**.

For the k' th filter stage, we must derive the polynomial coefficients a_{k1} and a_{k0} .

$$H(s) = \frac{1}{(s - s_1)(s - s_2) \cdots (s - s_n)}$$

$$H(s) = \frac{1}{(s - s_1) \underbrace{(s^2 + a_{11}s + a_{10})}_{\text{stage 1}} \underbrace{(s^2 + a_{21}s + a_{20})}_{\text{stage 2}} \cdots}$$

Odd order:

Mapping from linear pole indexing to complex conjugated pairs:

linear	paired
1,5	1,-1
2,4	2,-2
3	3

$$s_{k'} = e^{\frac{j\pi}{2n}(2k'+n-1)}, \quad k' = 1, 2, 3, \dots, n \quad \text{Linear indexing}$$

$$H(s) = \frac{1}{\left(s - e^{\frac{j\pi}{2n}\left(\frac{2(n+1)}{2}+n-1\right)}\right)} \cdot \prod_{k=1}^{\frac{n-1}{2}} \frac{1}{\left(s - e^{\frac{j\pi}{2n}(2k+n-1)}\right)\left(s - e^{\frac{j\pi}{2n}(2(n+1-k)+n-1)}\right)} \quad \text{Paired indexing}$$

$$H(s) = \frac{1}{(s - e^{j\pi})} \cdot \prod_{k=1}^{\frac{n-1}{2}} \frac{1}{\left(s - e^{\frac{j\pi}{2n}(2k+n-1)}\right)\left(s - e^{-\frac{j\pi}{2n}(2k+n-1)}\right)} \quad \text{Paired indexing}$$

$$H(s) = \frac{1}{s + 1} \cdot \prod_{k=1}^{\frac{n-1}{2}} \frac{1}{s^2 + a_{k1}s + a_{k0}} \quad \text{Paired indexing}$$

$$a_{k0} = e^{\frac{j\pi}{2n}(2k+n-1)} \cdot e^{-\frac{j\pi}{2n}(2k+n-1)} = 1$$

$$a_{k1} = -e^{\frac{j\pi}{2n}(2k+n-1)} - e^{-\frac{j\pi}{2n}(2k+n-1)} = -2 \cos\left(\frac{\pi}{2n}(2k + n - 1)\right)$$

Polynomial coefficients for Butterworth filters

Even order:

$$s_{k'} = e^{\frac{j\pi}{2n}(2k'+n-1)}, \quad k' = 1, 2, 3, \dots, n \quad \text{Linear indexing}$$

$$H(s) = \prod_{k=1}^{\frac{n}{2}} \frac{1}{\left(s - e^{\frac{j\pi}{2n}(2k+n-1)}\right) \left(s - e^{\frac{j\pi}{2n}(2(n+1-k)+n-1)}\right)} \quad \text{Paired indexing}$$

$$H(s) = \prod_{k=1}^{\frac{n}{2}} \frac{1}{\left(s - e^{\frac{j\pi}{2n}(2k+n-1)}\right) \left(s - e^{-\frac{j\pi}{2n}(2k+n-1)}\right)} \quad \text{Paired indexing}$$

$$H(s) = \prod_{k=1}^{\frac{n}{2}} \frac{1}{s^2 + a_{k1}s + a_{k0}} \quad \text{Paired indexing}$$

$$a_{k0} = e^{\frac{j\pi}{2n}(2k+n-1)} \cdot e^{-\frac{j\pi}{2n}(2k+n-1)} = 1$$

$$a_{k1} = -e^{\frac{j\pi}{2n}(2k+n-1)} - e^{-\frac{j\pi}{2n}(2k+n-1)} = -2 \cos\left(\frac{\pi}{2n}(2k+n-1)\right)$$

We are now able to tabulate the coefficients for each stage in a higher order Butterworth filter.

$$a_{k1} = -2 \cos\left(\frac{\pi}{2n}(2k + n - 1)\right)$$

$$a_{k0} = 1$$

For odd order filters, the 0'th stage is assumed to be the first order filter. For this stage $a_{k2} = 0$.

$$\omega_{nk} = \sqrt{a_{k0}} = 1$$

$$\zeta_k = \frac{a_{k1}}{2\sqrt{a_{k0}}} = \frac{a_{k1}}{2}$$

$$H(s) = \prod_{k=1}^L H_k(s) \quad H_k(s) = \frac{b_{k0}}{a_{k2}s^2 + a_{k1}s + a_{k0}}$$

n	k	a_{k2}	a_{k1}	a_{k0}
1	0	0.0	1.0	1.0
2	1	1.0	1.41421	1.0
3	0	0.0	1.0	1.0
	1	1.0	1.0	1.0
4	1	1.0	0.765367	1.0
	2	1.0	1.84776	1.0
5	0	0.0	1.0	1.0
	1	1.0	0.618034	1.0
	2	1.0	1.61803	1.0

Butterworth transfer functions

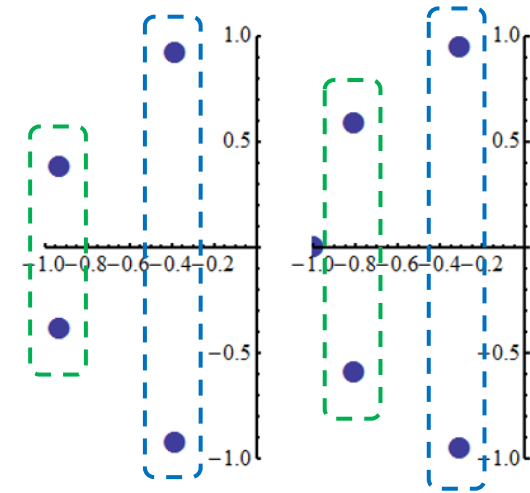
n	k	a_{k2}	a_{k1}	a_{k0}
1	0	0.0	1.0	1.0
2	1	1.0	1.41421	1.0
3	0	0.0	1.0	1.0
	1	1.0	1.0	1.0
4	1	1.0	0.765367	1.0
	2	1.0	1.84776	1.0
5	0	0.0	1.0	1.0
	1	1.0	0.618034	1.0
	2	1.0	1.61803	1.0

$$n = 2: \quad H_2(s) = \frac{1}{s^2 + 1.41421s + 1}$$

$$n = 3: \quad H_3(s) = \frac{1}{s + 1} \cdot \frac{1}{s^2 + 1.0s + 1}$$

$$n = 4: \quad H_4(s) = \underbrace{\frac{1}{s^2 + 0.765367s + 1}}_{\text{stage 1}} \cdot \underbrace{\frac{1}{s^2 + 1.84776s + 1}}_{\text{stage 2}}$$

$$n = 5: \quad H_5(s) = \frac{1}{s + 1} \cdot \underbrace{\frac{1}{s^2 + 0.618034s + 1}}_{\text{stage 1}} \cdot \underbrace{\frac{1}{s^2 + 1.61803s + 1}}_{\text{stage 2}}$$



Observation:

For the 4th and 5th order filters, the two quadratic factors have different coefficients. Hence the filter stages must have different component values. One 2nd order stage takes care of one pole pair, the other stage takes care of the other pole pair. **How do we see which pole pair belongs to which stage? Hint: ζ**

Butterworth filters

Matching to Sallen-Key circuit

Video

2nd order Sallen-Key low pass filter

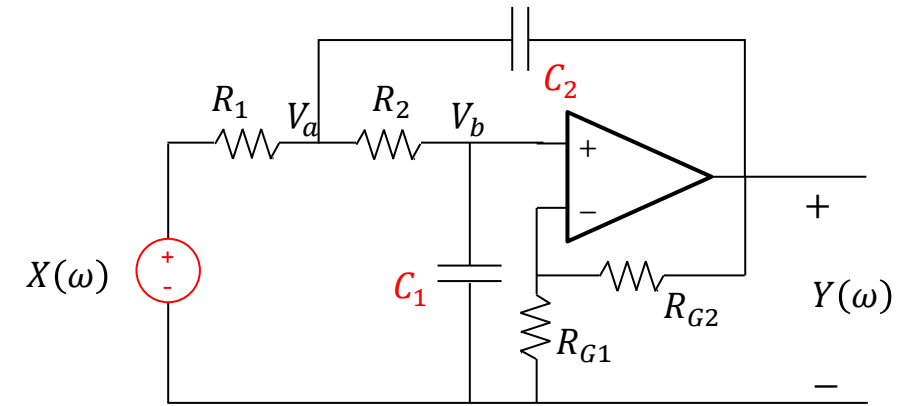
We will implement the Butterworth filter using a Sallen-Key lowpass circuit.

Be aware, that it is not the circuit that makes it a Butterworth filter, it is the **values of the coefficients** in the frequency characteristic.

By matching coefficients in the two equations, we obtain two equations:

$$R_1 R_2 C_1 C_2 = \frac{1}{a_0}$$

$$(R_2 C_1 + R_1 C_1 + R_1 C_2 (1 - K)) = \frac{a_1}{a_0}$$



$$H_{LP}(j\omega) = \frac{\frac{K}{R_1 R_2 C_1 C_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$H_{LP}(j\omega) = \frac{K}{R_1 R_2 C_1 C_2 (j\omega)^2 + j\omega (R_2 C_1 + R_1 C_1 + R_1 C_2 (1 - K)) + 1}$$

$$H_{LP}(j\omega) = \frac{K}{\frac{1}{a_0} (j\omega)^2 + \frac{a_1}{a_0} (j\omega) + 1}$$

2nd order Sallen-Key low pass filter

Isolating R_2 in the first equation and inserting it into the second, we obtain an equation for R_1 .

It is a quadratic equation.

We choose the positive solution:

We can then solve for R_2 :

$$R_2 = \frac{1}{a_0 R_1 C_1 C_2} \rightarrow (R_2 C_1 + R_1 C_1 + R_1 C_2 (1 - K)) = \frac{a_1}{a_0}$$

$$\frac{1}{a_0 R_1 C_2} + R_1 C_1 + R_1 C_2 (1 - K) = \frac{a_1}{a_0}$$

$$R_1 = \frac{a_1 - \sqrt{a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right)}}{2a_0 (C_1 + C_2 (1 - K))}$$

$$R_2 = \frac{a_1 + \sqrt{a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right)}}{2a_0 C_1}$$

For the resistors to have **real values**, we must ensure that the **discriminants are positive**.

$$R_1 = \frac{a_1 - \sqrt{a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right)}}{2a_0(C_1 + C_2(1 - K))}$$

$$R_2 = \frac{a_1 + \sqrt{a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right)}}{2a_0 C_1}$$

This puts a constraint on the capacitor values:

$$a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right) > 0 \Rightarrow \frac{a_1^2}{4a_0} > \frac{C_1}{C_2} + 1 - K \Rightarrow \frac{a_1^2}{4a_0} + K - 1 > \frac{C_1}{C_2}$$

Another constraint on the capacitor values ensures that the **resistors are positive**.

$$C_1 + C_2(1 - K) > 0 \Rightarrow C_1 > C_2(1 - K)$$

We can combine the two constraints:

$$C_2(K - 1) < C_1 < C_2 \left(\frac{a_1^2}{4a_0} + K - 1 \right)$$

So, if we first pick a value for C_2 , we can pick a value for C_1 within a bounded range.

Until now we have followed an approach that will provide a filter circuit that is realizable. At least we have avoided complex-valued or negative-valued resistors.

On the other hand, we have **not paid any attention to optimization** of some features of the system.

For electrical circuits, the uncertainty in performance associated with **variation in component values** is an important concern.

This variation can come about if a **malfunctioning component** must be replaced by a new component with a slightly different value, or because the **component value drifts** due to environmental factors (temperature, humidity, etc.).

We can apply **sensitivity analysis** to discover how sensitive a given circuit is to changes in component values.

$$C_2(K - 1) < C_1 < C_2 \left(\frac{a_1^2}{4a_0} + K - 1 \right)$$

$$R_1 = \frac{a_1 - \sqrt{a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right)}}{2a_0(C_1 + C_2(1 - K))}$$

$$R_2 = \frac{a_1 + \sqrt{a_1^2 - 4a_0 \left(\frac{C_1}{C_2} + 1 - K \right)}}{2a_0C_1}$$

Butterworth filters

Sensitivity of $H(j\omega)$ to drift in component values

Video

A second order transfer function is defined by two parameters, the resonance frequency ω_n and the damping ratio ζ .

In filter circuits, the **quality factor** Q is used instead of the damping ratio:

Inaccurate component values or large temperature drift can cause the real-life filter to have a frequency characteristic that differs from the intended. Different circuit topologies have different sensitivities.

A sensitivity analysis is called for in critical applications, and one circuit topology may be a better candidate than others because of lower sensitivity to **component drift**.

$$H_{LP}(j\omega) = \frac{K a_0}{(j\omega)^2 + j\omega a_1 + a_0}$$

$$H_{LP}(j\omega) = \frac{K \omega_n^2}{(j\omega)^2 + j\omega 2\zeta \omega_n + \omega_n^2}$$

Control theory

$$H_{LP}(j\omega) = \frac{K \omega_n^2}{(j\omega)^2 + j\omega \frac{\omega_n}{Q} + \omega_n^2}$$

Filter theory

$$\frac{1}{Q} \stackrel{\text{def}}{=} 2\zeta \Rightarrow Q \stackrel{\text{def}}{=} \frac{1}{2\zeta} = \frac{1}{2} \frac{2\sqrt{a_0}}{a_1} = \frac{\sqrt{a_0}}{a_1}$$

ζ	Q
0	∞
$1/\sqrt{2}$	$1/\sqrt{2}$
1	1/2

Sallen-Key frequency characteristic:

$$H_{LP}(j\omega) = \frac{\frac{K}{R_1 R_2 C_1 C_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

In sensitivity analysis, we can derive the sensitivity for $H(\omega)$ and obtain insight into what frequency range is most sensitive to change in component values.

$$\omega_n = (R_1 R_2 C_1 C_2)^{-1/2}$$

$$\frac{1}{Q} \stackrel{\text{def}}{=} 2\zeta \Rightarrow Q \stackrel{\text{def}}{=} \frac{1}{2\zeta} = \frac{1}{2} \frac{2\sqrt{a_0}}{a_1} = \frac{\sqrt{a_0}}{a_1}$$

We can also derive sensitivities for the circuit parameters, such as resonance frequency (ω_n) and the **Quality factor** (Q).

$$Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{R_2 C_1 + R_1 C_1 + (1-K)R_1 C_2}$$

To calculate the change in Q due to change in component values, we need expressions for Q :

$$\frac{1}{Q} = \sqrt{\frac{R_2 C_1}{R_1 C_2}} + \sqrt{\frac{R_1 C_1}{R_2 C_2}} + (1-K) \sqrt{\frac{R_1 C_2}{R_2 C_1}}$$

Sensitivity of a function y to a parameter x is defined as:

$$S_x^y \stackrel{\text{def}}{=} \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \lim_{\Delta x \rightarrow 0} \frac{x \Delta y}{y \Delta x} = \frac{x}{y} \frac{\partial y}{\partial x}$$

If $S_x^y = 5$, it implies that a 1% increase in x will produce a 5% *increase* in y .

If $S_x^y = -2$, it implies that a 1% increase in x will produce a 2% *decrease* in y .

Using this concept, we can analyse how many percent the amplitude and phase characteristics will change, if a component value is increased by 1%.

Our usual form:

$$H_{LP}(j\omega) = \frac{\frac{K}{R_1 R_2 C_1 C_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

An equivalent form:

$$H_{LP}(j\omega) = \frac{\frac{K}{R_1 R_2}}{(j\omega)^2 C_1 C_2 + j\omega \left(\frac{C_1}{R_1} + \frac{C_1}{R_2} + \frac{C_2(1-K)}{R_2} \right) + \frac{1}{R_1 R_2}} \stackrel{\text{def}}{=} \frac{N}{D}$$

Generic expression for the sensitivity function:

$$S_x^H = \frac{x}{H} \cdot \frac{\partial H}{\partial x} = x \left(\frac{N'}{N} - \frac{D'}{D} \right) \quad N' \stackrel{\text{def}}{=} \frac{\partial N}{\partial x}, D' \stackrel{\text{def}}{=} \frac{\partial D}{\partial x}$$

Using **polar form** in the sensitivity function:

$$S_x^H = \frac{x}{H} \cdot \frac{\partial H}{\partial x} = \frac{x}{|H(j\omega)| e^{j\theta(\omega)}} \left(\frac{\partial |H(j\omega)|}{\partial x} e^{j\theta(\omega)} + |H(j\omega)| \frac{\partial e^{j\theta(\omega)}}{\partial x} \right)$$

$$H(j\omega) = |H(j\omega)| e^{j\theta(\omega)}$$

Taking the derivative of a nested function:

$$S_x^H = \frac{x}{H} \cdot \frac{\partial H}{\partial x} = \frac{x}{|H(j\omega)| e^{j\theta(\omega)}} \left(\frac{\partial |H(j\omega)|}{\partial x} e^{j\theta(\omega)} + |H(j\omega)| e^{j\theta(\omega)} \frac{\partial (j\theta(\omega))}{\partial x} \right)$$

Multiplying into the parenthesis and simplifying:

$$S_x^H = \frac{x}{H} \cdot \frac{\partial H}{\partial x} = \frac{x}{|H(j\omega)|e^{j\theta(\omega)}} \left(\frac{\partial |H(j\omega)|}{\partial x} e^{j\theta(\omega)} + |H(j\omega)| e^{j\theta(\omega)} \frac{\partial (j\theta(\omega))}{\partial x} \right)$$

$$S_x^H = \frac{x}{H} \cdot \frac{\partial H}{\partial x} = \underbrace{\frac{x}{|H(j\omega)|} \frac{\partial |H(j\omega)|}{\partial x}}_{S_x^{|H|}} + j|\theta| \underbrace{\frac{x}{|\theta|} \frac{\partial |\theta|(\omega)}{\partial x}}_{S_x^{|\theta|}}$$

$$S_x^H = S_x^{|H|} + j|\theta| S_x^{|\theta|} = x \left(\frac{N'}{N} - \frac{D'}{D} \right)$$

$$S_x^{|H|} = \operatorname{Re} \left\{ x \left(\frac{N'}{N} - \frac{D'}{D} \right) \right\} \quad \text{Magnitude sensitivity}$$

$$S_x^{|\theta|} = \frac{1}{|\theta|} \operatorname{Im} \left\{ x \left(\frac{N'}{N} - \frac{D'}{D} \right) \right\} \quad \text{Phase sensitivity}$$

The sensitivity of $H(j\omega)$ to a parameter will be a complex-valued function of ω .

We see that the real part of the derived sensitivity is the sensitivity of the magnitude function.

The imaginary part divided by the phase (in radians) is the sensitivity function for the phase.

The sensitivity to change in C_1

$$S_{C_1}^H = \frac{\omega^2 C_1 C_2 - j\omega C_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right)}{D} = \begin{cases} 0 & \omega \rightarrow 0 \\ -1 & \omega \rightarrow \infty \end{cases}$$

All expressions have the same denominator:

$$D = (j\omega)^2 C_1 C_2 + j\omega \left(\frac{C_1}{R_1} + \frac{C_1}{R_2} + \frac{C_2(1-K)}{R_2} \right) + \frac{1}{R_1 R_2}$$

The sensitivity to change in C_2

$$S_{C_2}^H = \frac{\omega^2 C_1 C_2 - j\omega \frac{C_2}{R_2} (1-K)}{D} = \begin{cases} 0 & \omega \rightarrow 0 \\ -1 & \omega \rightarrow \infty \end{cases}$$

The sensitivity to change in R_1

$$S_{R_1}^H = \frac{\omega^2 C_1 C_2 - j\omega \left(\frac{C_1}{R_2} + \frac{C_2}{R_2} (1-K) \right)}{D} = \begin{cases} 0 & \omega \rightarrow 0 \\ -1 & \omega \rightarrow \infty \end{cases}$$

The sensitivity to change in R_2

$$S_{R_2}^H = \frac{\omega^2 C_1 C_2 - j\omega \frac{C_1}{R_1}}{D} = \begin{cases} 0 & \omega \rightarrow 0 \\ -1 & \omega \rightarrow \infty \end{cases}$$

The sensitivity to change in K

$$S_K^H = \frac{-\omega^2 C_1 C_2 + j\omega \left(\frac{C_1}{R_2} + \frac{C_2}{R_2} + \frac{C_1}{R_1} \right) + \frac{1}{R_1 R_2}}{D} = \begin{cases} 1 & \omega \rightarrow 0 \\ 1 & \omega \rightarrow \infty \end{cases}$$

Implementation in Maple.

We define a numerator function and a denominator function:

Then we define a general sensitivity function to which we can pass any component symbol

$$(x = R_1, R_2, C_1, C_2, K)$$

Then we call the function, passing C_1 , and obtains an expression for the sensitivity to C_1 . This is a function of ω and can be plotted for a range of frequencies, if we know the values of the components.

$$H := \omega \rightarrow \frac{\frac{K}{R1 \cdot R2}}{(j \omega)^2 \cdot C1 \cdot C2 + (j \omega) \cdot \left(\frac{C1}{R2} + \frac{C1}{R1} + \frac{C2 \cdot (1 - K)}{R2} \right) + \frac{1}{R1 \cdot R2}} :$$

$$Den := \omega \rightarrow (j \omega)^2 \cdot C1 \cdot C2 + (j \omega) \cdot \left(\frac{C1}{R2} + \frac{C1}{R1} + \frac{C2 \cdot (1 - K)}{R2} \right) + \frac{1}{R1 \cdot R2} :$$

$$Num := \omega \rightarrow \frac{K}{R1 \cdot R2} :$$

$$SH := (x, \omega) \rightarrow x \cdot \left(\frac{\left(\frac{\partial}{\partial x} (Num(\omega)) \right)}{Num(\omega)} - \frac{\left(\frac{\partial}{\partial x} (Den(\omega)) \right)}{Den(\omega)} \right) :$$

$$SH(C1, \omega)$$

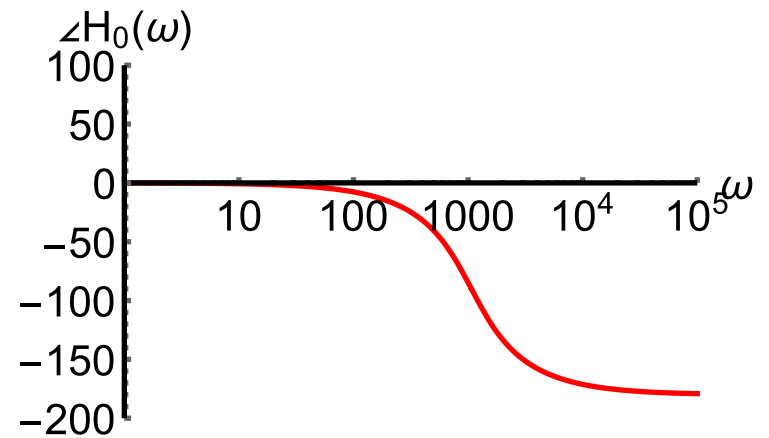
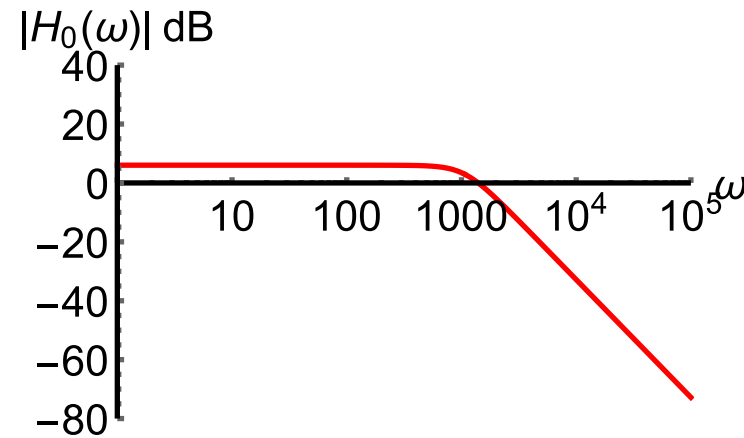
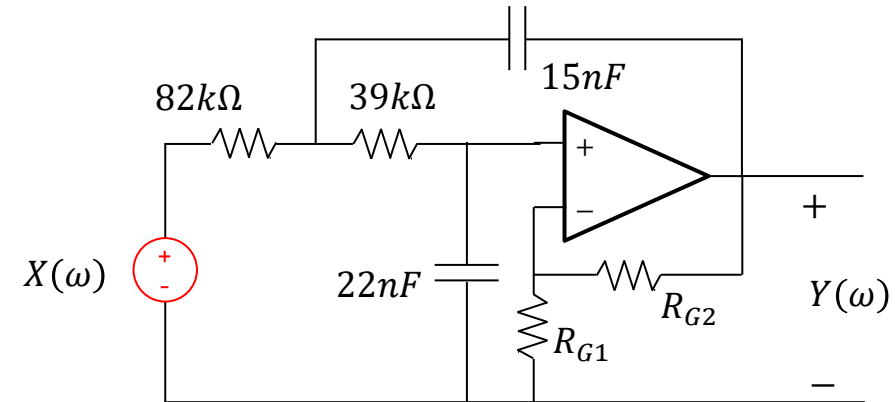
$$- \frac{C1 \left(-\omega^2 C2 + I \omega \left(\frac{1}{R2} + \frac{1}{R1} \right) \right)}{-\omega^2 C1 C2 + I \omega \left(\frac{C1}{R2} + \frac{C1}{R1} + \frac{C2 (1 - K)}{R2} \right) + \frac{1}{R1 R2}}$$

$$SHC1 := \omega \rightarrow - \frac{C1 \left(-\omega^2 C2 + I \omega \left(\frac{1}{R2} + \frac{1}{R1} \right) \right)}{-\omega^2 C1 C2 + I \omega \left(\frac{C1}{R2} + \frac{C1}{R1} + \frac{C2 (1 - K)}{R2} \right) + \frac{1}{R1 R2}} :$$

Example: 2nd order Sallen-Key lowpass filter

Example lowpass filter.

We will use these component values to analyse the sensitivity of the frequency characteristics to the components.



To plot the sensitivity of the *magnitude* we just take the **real part of $S_x^H(\omega)$** .

```
Sp1 := plot([Re(SHC1(ω)), Re(SHC2(ω)), Re(SHR1(ω)), Re(SHR2(ω)), Re(SHK(ω))], ω = 1..4E3,
-3..3, color = ["Red", "Brown", "Blue", "Green", "Black"], thickness = 3, font = [Helvetica, roman, 18],
axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = ["ω, rad/s", typeset(Sx|H|)], labelfont
= [Helvetica, 18], numpoints = 100, gridlines, size = [600, 200]) :
```

To plot the sensitivity of the *phase* we take **the imaginary part of $S_x^H(\omega)$ and divide by the absolute value of the phase in radians**.

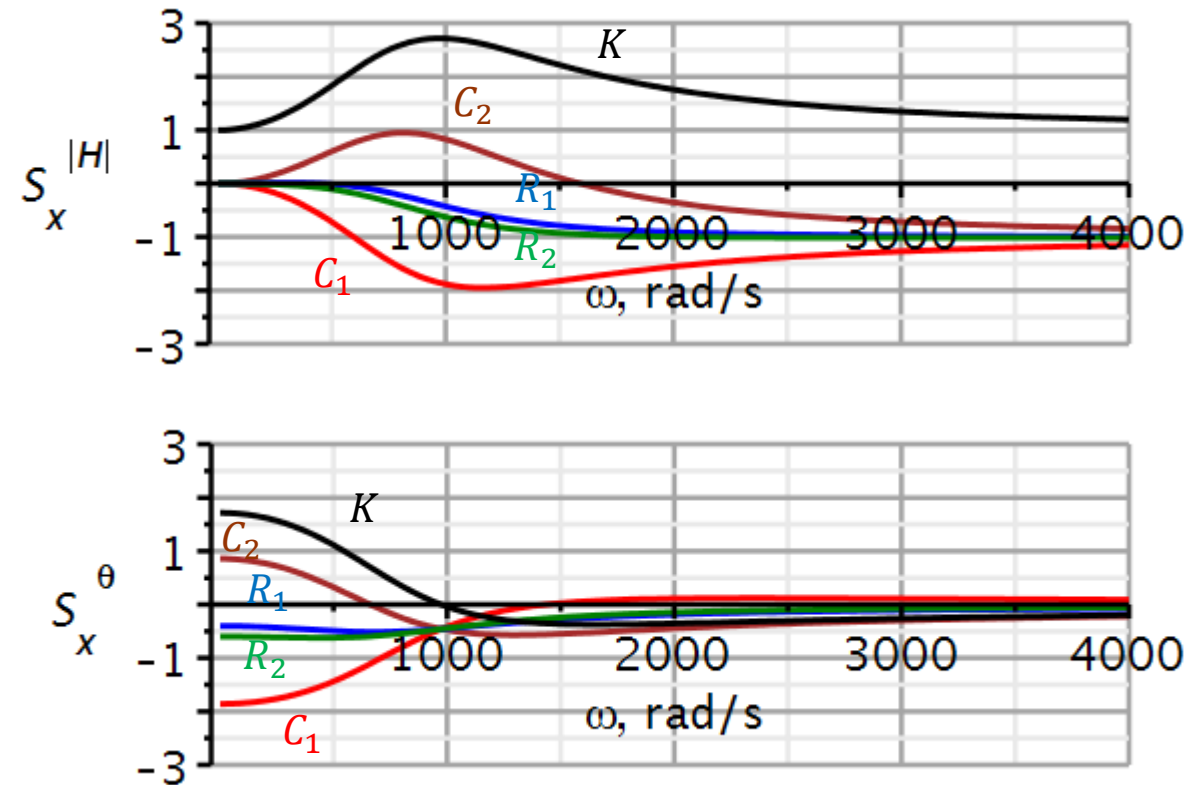
```
Sp2 := plot([ [ Im(SHC1(ω)) / |radian(H(ω))|, Im(SHC2(ω)) / |radian(H(ω))|, Im(SHR1(ω)) / |radian(H(ω))|, Im(SHR2(ω)) / |radian(H(ω))|,
Im(SHK(ω)) / |radian(H(ω))| ], ω = 1..4E3, -3..3, color = ["Red", "Brown", "Blue", "Green", "Black"], thickness = 3,
font = [Helvetica, roman, 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels
= ["ω, rad/s", typeset(Sxθ)], labelfont = [Helvetica, 18], numpoints = 100, gridlines, size = [600, 200]) :
plotstack := Array(1..2, 1..1) :
plotstack[1, 1] := Sp1 :
plotstack[2, 1] := Sp2 :
display(plotstack, aligncolumns = [1])
```

We see, that the largest sensitivity is to the gain K . A 1% increase in K will increase the magnitude by almost 3% at the 3dB cut-off frequency.

This suggest that a unity gain filter would be better since in that case there are no gain resistors that can drift.

We also notice that similar changes in C_1 and C_2 will change the amplitude in opposite directions, thus tending to cancel out. This is not the case for similar drift in the two resistors.

We confirm that the magnitude sensitivity agrees with the low- and high-frequency asymptotes derived previously.



Butterworth filters

Sensitivity of Q and ω_n

Video

Now we will investigate the sensitivity of the resonance frequency ω_n and the quality factor Q .

$$H_{LP}(j\omega) = \frac{\frac{K}{R_1 R_2 C_1 C_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$H_{LP}(j\omega) = \frac{\frac{K}{\omega_n^2}}{(j\omega)^2 + j\omega \frac{\omega_n}{Q} + \omega_n^2}$$

$$\omega_n = (R_1 R_2 C_1 C_2)^{-1/2}$$

$$\frac{1}{Q} \stackrel{\text{def}}{=} 2\zeta \Rightarrow Q \stackrel{\text{def}}{=} \frac{1}{2\zeta} = \frac{1}{2} \frac{2\sqrt{a_0}}{a_1} = \frac{\sqrt{a_0}}{a_1}$$

$$Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{R_2 C_1 + R_1 C_1 + (1-K)R_1 C_2}$$

Using Maple, we can obtain sensitivity functions:

$$S_x^y \stackrel{\text{def}}{=} \frac{x}{y} \frac{\partial y}{\partial x}$$

Observation:

All components have the same impact on the resonance frequency, and the **component values are irrelevant**.

DC gain K has no impact on the resonance frequency.

The sensitivity of the Q factor is component dependent and needs further investigation.

$$S_{R_1}^{\omega_n} = -\frac{1}{2}$$

$$S_{R_2}^{\omega_n} = -\frac{1}{2}$$

$$S_{C_1}^{\omega_n} = -\frac{1}{2}$$

$$S_{C_2}^{\omega_n} = -\frac{1}{2}$$

$$S_K^{\omega_n} = 0$$

$$S_{R_1}^Q = \frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 - R_1)}{(K-1)R_1C_2 + C_1(R_1 + R_2)}$$

$$S_{R_2}^Q = -\frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 - R_1)}{(K-1)R_1C_2 + C_1(R_1 + R_2)}$$

$$S_{C_1}^Q = \frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 + R_1)}{(K-1)R_1C_2 - C_1(R_1 + R_2)}$$

$$S_{C_2}^Q = -\frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 + R_1)}{(K-1)R_1C_2 - C_1(R_1 + R_2)}$$

$$S_K^Q = \frac{K R_1 C_2}{(K-1)R_1C_2 + (R_1 + R_2)C_1}$$

$$\omega_n := \sqrt{\frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}} :$$

$$Q := \frac{\sqrt{R1 \cdot R2 \cdot C1 \cdot C2}}{C1 \cdot (R1 + R2) + R1 \cdot C2 \cdot (1 - K)} :$$

$$S(y, x) := \frac{x}{y} \frac{\partial}{\partial x}(y) :$$

$$S(\omega_n, R1) \quad -\frac{1}{2}$$

$$S(\omega_n, R2) \quad -\frac{1}{2}$$

$$S(\omega_n, C1) \quad -\frac{1}{2}$$

$$S(\omega_n, C2) \quad -\frac{1}{2}$$

$$SQR1 := \text{simplify}(S(Q, R1))$$

$$SQR1 := \frac{(C1 + C2 (1 - K)) R1 - R2 C1}{(-2 C1 + (2 K - 2) C2) R1 - 2 R2 C1}$$

$$SQR2 := \text{simplify}(S(Q, R2))$$

$$SQR2 := \frac{(-C1 + (K - 1) C2) R1 + R2 C1}{(-2 C1 + (2 K - 2) C2) R1 - 2 R2 C1}$$

$$SQC1 := \text{simplify}(S(Q, C1))$$

$$SQC1 := \frac{(C1 + (K - 1) C2) R1 + R2 C1}{(-2 C1 + (2 K - 2) C2) R1 - 2 R2 C1}$$

$$SQC2 := \text{simplify}(S(Q, C2))$$

$$SQC2 := \frac{(-C1 + C2 (1 - K)) R1 - R2 C1}{(-2 C1 + (2 K - 2) C2) R1 - 2 R2 C1}$$

$$SQK := \text{simplify}(S(Q, K))$$

$$SQK := -\frac{K C2 R1}{(-C1 + (K - 1) C2) R1 - R2 C1}$$

From our analysis of the sensitivity of the frequency characteristics, we observed that the sensitivity to gain is the largest. **Hence, we would like the gain to be unity, so that it is independent of component values.**

$$K \stackrel{\text{def}}{=} 1$$

Observations:

The sensitivities to the two resistors have opposite sign. The same is the case for the capacitors. If they drift equally, their drift cancels out.

If $R_1 \stackrel{\text{def}}{=} R_2$, then the sensitivity of Q to the resistors becomes zero.

$$S_{R_1}^Q = \frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 - R_1)}{(K-1)R_1C_2 + C_1(R_1 + R_2)} = \frac{1}{2} \cdot \frac{(R_2 - R_1)}{(R_1 + R_2)}$$

$$S_{R_2}^Q = -\frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 - R_1)}{(K-1)R_1C_2 + C_1(R_1 + R_2)} = -\frac{1}{2} \cdot \frac{(R_2 - R_1)}{(R_1 + R_2)}$$

$$S_{C_1}^Q = \frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 + R_1)}{(K-1)R_1C_2 - C_1(R_1 + R_2)} = -\frac{1}{2} \cdot \frac{(R_2 + R_1)}{(R_1 + R_2)} = -\frac{1}{2}$$

$$S_{C_2}^Q = -\frac{1}{2} \cdot \frac{(K-1)R_1C_2 + C_1(R_2 + R_1)}{(K-1)R_1C_2 - C_1(R_1 + R_2)} = \frac{1}{2} \cdot \frac{(R_2 + R_1)}{(R_1 + R_2)} = \frac{1}{2}$$

$$S_K^Q = \frac{K R_1 C_2}{(K-1)R_1C_2 + (R_1 + R_2)C_1} = \frac{K R_1 C_2}{(R_1 + R_2)C_1}$$

So, for the present filter it is a good idea to match resistor values.

For this particular circuit:

- Matching resistor values will nullify the sensitivity of Q to the resistors.
- Matching capacitor values will not have the same effect.

We will match resistor values, when choosing component values.

Observation:

We are analysing the Sallen-Key lowpass filter. Our conclusions only apply to this circuit.

For a Sallen-Key highpass filter, our conclusions will be different.

$$K = 1, R_1 = R_2$$

$$S_{R_1}^Q = \frac{1}{2} \cdot \frac{(R_2 - R_1)}{(R_1 + R_2)} = 0$$

$$S_{R_2}^Q = -\frac{1}{2} \cdot \frac{(R_2 - R_1)}{(R_1 + R_2)} = 0$$

$$S_{C_1}^Q = -\frac{1}{2}$$

$$S_{C_2}^Q = \frac{1}{2}$$

Butterworth filters

Sensitivity-optimized filter design

Video

We will first calculate component values for a frequency-normalized filter,

$$\omega_n = 1 \text{ rad/s.}$$

We will use what we have learned from the sensitivity analysis to design the Sallen-Key lowpass filter.

$$K = 1; R_1 = R_2$$

Combining these two expressions in such a way that R_1 cancel gives us a break-through:

$$H_{LP}(j\hat{\omega}) = \frac{\frac{K}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}{(j\hat{\omega})^2 + j\hat{\omega} \left(\frac{1}{\hat{R}_1 \hat{C}_2} + \frac{1}{\hat{R}_2 \hat{C}_2} + \frac{(1-K)}{\hat{R}_2 \hat{C}_1} \right) + \frac{1}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}$$

$$H_{LP}(j\hat{\omega}) = \frac{\frac{1}{\hat{R}_1^2 \hat{C}_1 \hat{C}_2}}{(j\hat{\omega})^2 + j\hat{\omega} \left(\underbrace{\frac{2}{\hat{R}_1 \hat{C}_2}}_{a_1} \right) + \underbrace{\frac{1}{\hat{R}_1^2 \hat{C}_1 \hat{C}_2}}_{a_0}}$$

$$a_1 = \frac{2}{\hat{R}_1 \hat{C}_2} \quad a_0 = \frac{1}{\hat{R}_1^2 \hat{C}_1 \hat{C}_2}$$

$$\frac{4a_0}{a_1^2} = \frac{\frac{4}{\hat{R}_1^2 \hat{C}_1 \hat{C}_2}}{\frac{4}{(\hat{R}_1 \hat{C}_2)^2}} = \frac{\hat{C}_2}{\hat{C}_1} \Rightarrow \quad \frac{\hat{C}_2}{\hat{C}_1} = \frac{4a_0}{a_1^2} \quad \hat{R}_2 = \hat{R}_1 = \frac{2}{a_1 \hat{C}_2}$$

Example: $a_0 = 1$
 $a_1 = \sqrt{2}$

We need to find capacitor values to obtain a specific ratio. We can search a table (next slide) for possible ratios and choose the best ratio.

Having decided on $\hat{C}_1$, the product $\hat{R}_1\hat{C}_1$ must have a certain value.

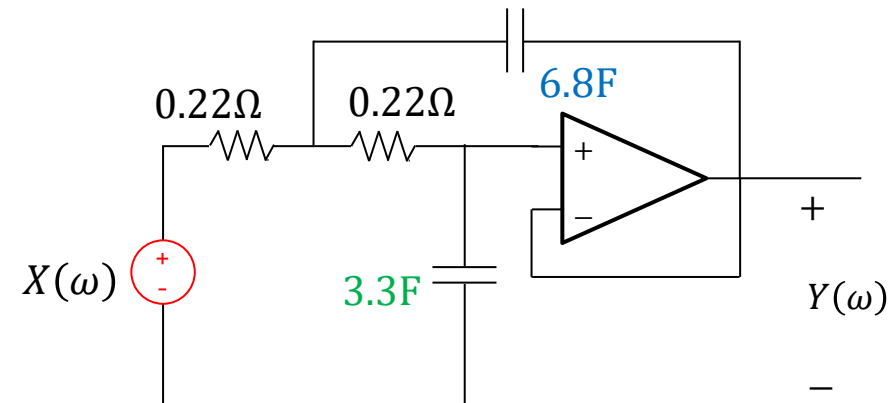
The best value can be looked up in a table (2nd next slide).

$$\frac{\hat{C}_2}{\hat{C}_1} = \frac{4a_0}{a_1^2} = \frac{4 \cdot 1}{2} = 2 \approx \frac{6.8}{3.3} \quad \text{From a look-up table, next slide}$$

$$\hat{R}_1\hat{C}_1 = \hat{R}_2\hat{C}_1 = \frac{a_1}{2a_0} = \frac{\sqrt{2}}{2 \cdot 1} = \frac{1}{\sqrt{2}} = 0.707 \Rightarrow \hat{R}_1 = \frac{0.707}{3.3} = 0.214$$

Our choice of component values for the normalized filter.

$$\hat{C}_1 = 3.3\text{F}, \hat{C}_2 = 6.8\text{F}, \hat{R}_1 = 0.22\Omega, \hat{R}_2 = 0.22\Omega$$





Design procedure for low sensitivity

$$\frac{\hat{C}_2}{\hat{C}_1} = \frac{4a_0}{a_1^2} = \frac{4 \cdot 1}{2} = 2 \approx \frac{6.8}{3.3}$$

$C_2 \rightarrow$

C2/C1	1	1,2	1,5	1,8	2,2	2,7	3,3	3,9	4,7	5,6	6,8	8,2	10	12	15	18	22	27	33	39	47	56	68	82	100
1	1,0000	1,2000	1,5000	1,8000	2,2000	2,7000	3,3000	3,9000	4,7000	5,6000	6,8000	8,2000	10,0000	12,0000	15,0000	18,0000	22,0000	27,0000	33,0000	39,0000	47,0000	56,0000	68,0000	82,0000	100,0000
1,2	0,8333	1,0000	1,2500	1,5000	1,8333	2,2500	2,7500	3,2500	3,9167	4,6667	5,6667	6,8333	8,3333	10,0000	12,5000	15,0000	18,3333	22,5000	27,5000	32,5000	39,1667	46,6667	56,6667	68,3333	83,3333
1,5	0,6667	0,8000	1,0000	1,2000	1,4667	1,8000	2,2000	2,6000	3,1333	3,7333	4,5333	5,4667	6,6667	8,0000	10,0000	12,0000	14,6667	18,0000	22,0000	26,0000	31,3333	37,3333	45,3333	54,6667	66,6667
1,8	0,5556	0,6667	0,8333	1,0000	1,2222	1,5000	1,8333	2,1667	2,6111	3,1111	3,7778	4,5556	5,5556	6,6667	8,3333	10,0000	12,2222	15,0000	18,3333	21,6667	26,1111	31,1111	37,7778	45,5556	55,5556
2,2	0,4545	0,5455	0,6818	0,8182	1,0000	1,2273	1,5000	1,7727	2,1364	2,5455	3,0909	3,7273	4,5455	5,4545	6,8182	8,1818	10,0000	12,2727	15,0000	17,7273	21,3636	25,4545	30,9091	37,2727	45,4545
2,7	0,3704	0,4444	0,5556	0,6667	0,8148	1,0000	1,2222	1,4444	1,7407	2,0741	2,5185	3,0370	3,7037	4,4444	5,5556	6,6667	8,1481	10,0000	12,2222	14,4444	17,4074	20,7407	25,1852	30,3704	37,0370
3,3	0,3030	0,3636	0,4545	0,5455	0,6667	0,8182	1,0000	1,1818	1,4242	1,6970	2,0606	2,4848	3,0303	3,6364	4,5455	5,4545	6,6667	8,1818	10,0000	11,8182	14,2424	16,9697	20,6061	24,8485	30,3030
3,9	0,2564	0,3077	0,3846	0,4615	0,5641	0,6923	0,8462	1,0000	1,2051	1,4359	1,7436	2,1026	2,5641	3,0769	3,8462	4,6154	5,6410	6,9231	8,4615	10,0000	12,0513	14,3590	17,4359	21,0256	25,6410
4,7	0,2128	0,2553	0,3191	0,3830	0,4681	0,5745	0,7021	0,8298	1,0000	1,1915	1,4468	1,7447	2,1277	2,5532	3,1915	3,8298	4,6809	5,7447	7,0213	8,2979	10,0000	11,9149	14,4681	17,4468	21,2766
5,6	0,1786	0,2143	0,2679	0,3214	0,3929	0,4821	0,5893	0,6964	0,8393	1,0000	1,2143	1,4643	1,7857	2,1429	2,6786	3,2143	3,9286	4,8214	5,8929	6,9643	8,3929	10,0000	12,1429	14,6429	17,8571
6,8	0,1471	0,1765	0,2206	0,2647	0,3235	0,3971	0,4853	0,5735	0,6912	0,8235	1,0000	1,2059	1,4706	1,7647	2,2059	2,6471	3,2353	3,9706	4,8529	5,7353	6,9118	8,2353	10,0000	12,0588	14,7059
8,2	0,1220	0,1463	0,1829	0,2195	0,2683	0,3293	0,4024	0,4756	0,5732	0,6829	0,8293	1,0000	1,2195	1,4634	1,8293	2,1951	2,6829	3,2927	4,0244	4,7561	5,7317	6,8293	8,2927	10,0000	12,1951
10	0,1000	0,1200	0,1500	0,1800	0,2200	0,2700	0,3300	0,3900	0,4700	0,5600	0,6800	0,8200	1,0000	1,2000	1,5000	1,8000	2,2000	2,7000	3,3000	3,9000	4,7000	5,6000	6,8000	8,2000	10,0000
12	0,0833	0,1000	0,1250	0,1500	0,1833	0,2250	0,2750	0,3250	0,3917	0,4667	0,5667	0,6833	0,8333	1,0000	1,2500	1,5000	1,8333	2,2500	2,7500	3,2500	3,9167	4,6667	5,6667	6,8333	8,3333
15	0,0667	0,0800	0,1000	0,1200	0,1467	0,1800	0,2200	0,2600	0,3133	0,3733	0,4533	0,5467	0,6667	0,8000	1,0000	1,2000	1,4667	1,8000	2,2000	2,6000	3,1333	3,7333	4,5333	5,4667	6,6667
18	0,0556	0,0667	0,0833	0,1000	0,1222	0,1500	0,1833	0,2167	0,2611	0,3111	0,3778	0,4556	0,5556	0,6667	0,8333	1,0000	1,2222	1,5000	1,8333	2,1667	2,6111	3,1111	3,7778	4,5556	5,5556
22	0,0455	0,0545	0,0682	0,0818	0,1000	0,1227	0,1500	0,1773	0,2136	0,2545	0,3091	0,3727	0,4545	0,5455	0,6818	0,8182	1,0000	1,2273	1,5000	1,7727	2,1364	2,5455	3,0909	3,7273	4,5455
27	0,0370	0,0444	0,0556	0,0667	0,0815	0,1000	0,1222	0,1444	0,1741	0,2074	0,2519	0,3037	0,3704	0,4444	0,5556	0,6667	0,8148	1,0000	1,2222	1,4444	1,7407	2,0741	2,5185	3,0370	3,7037
33	0,0303	0,0364	0,0455	0,0545	0,0667	0,0818	0,1000	0,1182	0,1424	0,1697	0,2061	0,2485	0,3030	0,3636	0,4545	0,5455	0,6667	0,8182	1,0000	1,1818	1,4242	1,6970	2,0606	2,4848	3,0303
39	0,0256	0,0308	0,0385	0,0462	0,0564	0,0692	0,0846	0,1000	0,1205	0,1436	0,1744	0,2103	0,2564	0,3077	0,3846	0,4615	0,5641	0,6923	0,8462	1,0000	1,2051	1,4359	1,7436	2,1026	2,5641
47	0,0213	0,0255	0,0319	0,0383	0,0468	0,0574	0,0702	0,0830	0,1000	0,1191	0,1447	0,1745	0,2128	0,2553	0,3191	0,3830	0,4681	0,5745	0,7021	0,8298	1,0000	1,1915	1,4468	1,7447	2,1277
56	0,0179	0,0214	0,0268	0,0321	0,0393	0,0482	0,0589	0,0696	0,0839	0,1000	0,1214	0,1464	0,1786	0,2143	0,2679	0,3214	0,3929	0,4821	0,5893	0,6964	0,8393	1,0000	1,2143	1,4643	1,7857
68	0,0147	0,0176	0,0221	0,0265	0,0324	0,0397	0,0485	0,0574	0,0691	0,0824	0,1000	0,1206	0,1471	0,1765	0,2206	0,2647	0,3235	0,3971	0,4853	0,5735	0,6912	0,8235	1,0000	1,2059	1,4706
82	0,0122	0,0146	0,0183	0,0220	0,0268	0,0329	0,0402	0,0476	0,0573	0,0683	0,0829	0,1000	0,1220	0,1463	0,1829	0,2195	0,2683	0,3293	0,4024	0,4756	0,5732	0,6829	0,8293	1,0000	1,2195
100	0,0100	0,0120	0,0150	0,0180	0,0220	0,0270	0,0330	0,0390	0,0470	0,0560	0,0680	0,0820	0,1000	0,1200	0,1500	0,1800	0,2200	0,2700	0,3300	0,3900	0,4700	0,5600	0,6800	0,8200	1,0000

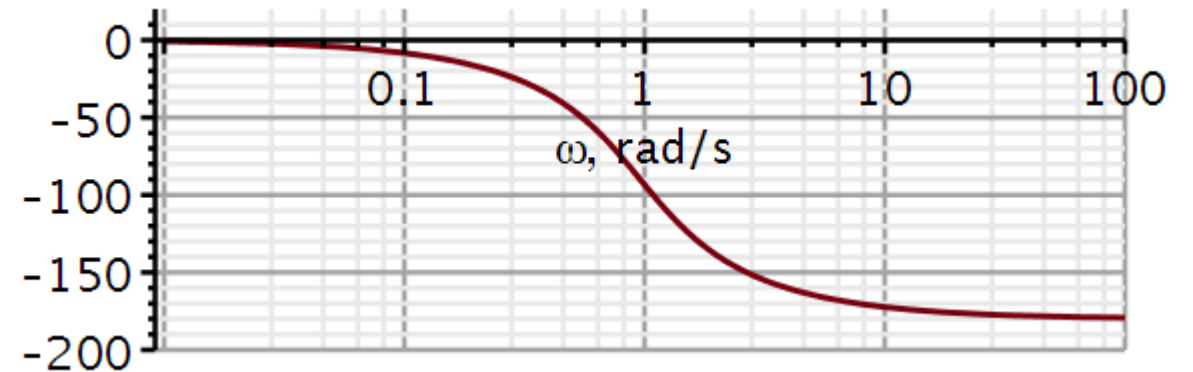
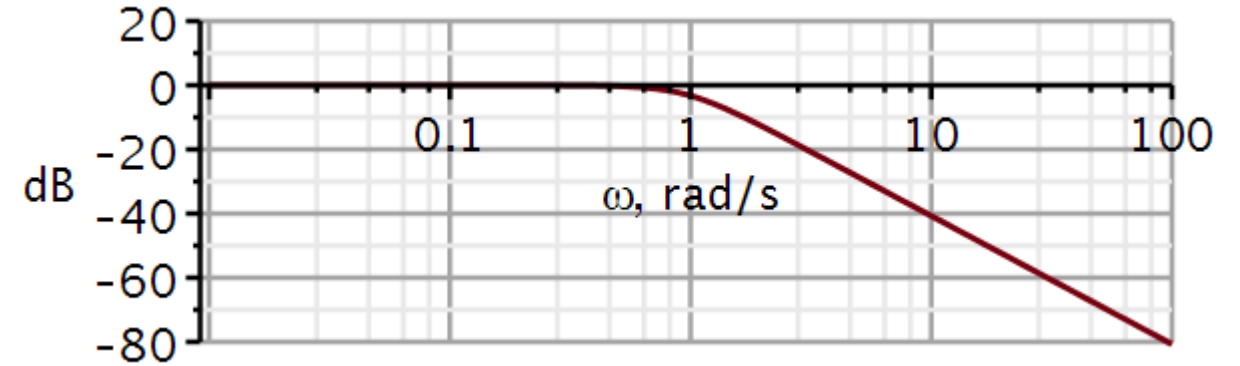
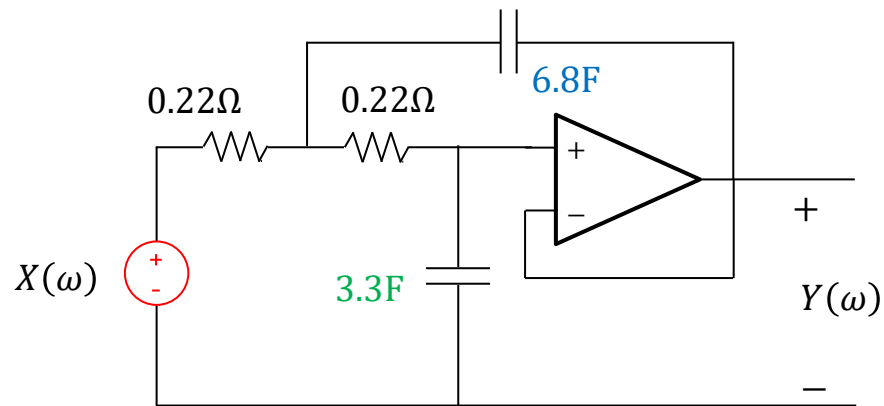
Design procedure for low sensitivity

$$\hat{R}_1 \cdot \hat{C}_1 = \frac{a_1}{2a_0} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} = 0.7071$$



R1*C1	0,1	0,12	0,15	0,18	0,22	0,27	0,33	0,39	0,47	0,56	0,68	0,82	1	1,2	1,5	1,8	2,2	2,7	3,3	3,9	4,7	5,6	6,8	8,2	10
0,1	0,010	0,012	0,015	0,018	0,022	0,027	0,033	0,039	0,047	0,056	0,068	0,082	0,100	0,120	0,150	0,180	0,220	0,270	0,330	0,390	0,470	0,560	0,680	0,820	1,000
0,12	0,012	0,014	0,018	0,022	0,026	0,032	0,040	0,047	0,056	0,067	0,082	0,098	0,120	0,144	0,180	0,216	0,264	0,324	0,396	0,468	0,564	0,672	0,816	0,984	1,200
0,15	0,015	0,018	0,023	0,027	0,033	0,041	0,050	0,059	0,071	0,084	0,102	0,123	0,150	0,180	0,225	0,270	0,330	0,405	0,495	0,585	0,705	0,840	1,020	1,230	1,500
0,18	0,018	0,022	0,027	0,032	0,040	0,049	0,059	0,070	0,085	0,101	0,122	0,148	0,180	0,216	0,270	0,324	0,396	0,486	0,594	0,702	0,846	1,008	1,224	1,476	1,800
0,22	0,022	0,026	0,033	0,040	0,048	0,059	0,073	0,086	0,103	0,123	0,150	0,180	0,220	0,264	0,330	0,396	0,484	0,594	0,726	0,858	1,034	1,232	1,496	1,804	2,200
0,27	0,027	0,032	0,041	0,049	0,059	0,073	0,089	0,105	0,127	0,151	0,184	0,221	0,270	0,324	0,405	0,486	0,594	0,729	0,891	1,053	1,269	1,512	1,836	2,214	2,700
0,33	0,033	0,040	0,050	0,059	0,073	0,089	0,109	0,129	0,155	0,185	0,224	0,271	0,330	0,396	0,495	0,594	0,726	0,891	1,089	1,287	1,551	1,848	2,244	2,706	3,300
0,39	0,039	0,047	0,059	0,070	0,086	0,105	0,129	0,152	0,183	0,218	0,265	0,320	0,390	0,468	0,585	0,702	0,858	1,053	1,287	1,521	1,833	2,184	2,652	3,198	3,900
0,47	0,047	0,056	0,071	0,085	0,103	0,127	0,155	0,183	0,221	0,263	0,320	0,385	0,470	0,564	0,705	0,846	1,034	1,269	1,551	1,833	2,209	2,632	3,196	3,854	4,700
0,56	0,056	0,067	0,084	0,101	0,123	0,151	0,185	0,218	0,263	0,314	0,381	0,459	0,560	0,672	0,840	1,008	1,232	1,512	1,848	2,184	2,632	3,136	3,808	4,592	5,600
0,68	0,068	0,082	0,102	0,122	0,150	0,184	0,224	0,265	0,320	0,381	0,462	0,558	0,680	0,816	1,020	1,224	1,496	1,836	2,244	2,652	3,196	3,808	4,624	5,576	6,800
0,82	0,082	0,098	0,123	0,148	0,180	0,221	0,271	0,320	0,385	0,459	0,558	0,672	0,820	0,984	1,230	1,476	1,804	2,214	2,706	3,198	3,854	4,592	5,576	6,724	8,200
1	0,100	0,120	0,150	0,180	0,220	0,270	0,330	0,390	0,470	0,560	0,680	0,820	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000
1,2	0,120	0,144	0,180	0,216	0,264	0,324	0,396	0,468	0,564	0,672	0,816	0,984	1,200	1,440	1,800	2,160	2,640	3,240	3,960	4,680	5,640	6,720	8,160	9,840	12,000
1,5	0,150	0,180	0,225	0,270	0,330	0,405	0,495	0,585	0,705	0,840	1,020	1,230	1,500	1,800	2,250	2,700	3,300	4,050	4,950	5,850	7,050	8,400	10,200	12,300	15,000
1,8	0,180	0,216	0,270	0,324	0,396	0,486	0,594	0,702	0,846	1,008	1,224	1,476	1,800	2,160	2,700	3,240	3,960	4,860	5,940	7,020	8,460	10,080	12,240	14,760	18,000
2,2	0,220	0,264	0,330	0,396	0,484	0,594	0,726	0,858	1,034	1,232	1,496	1,804	2,200	2,640	3,300	3,960	4,840	5,940	7,260	8,580	10,340	12,320	14,960	18,040	22,000
2,7	0,270	0,324	0,405	0,486	0,594	0,729	0,891	1,053	1,269	1,512	1,836	2,214	2,700	3,240	4,050	4,860	5,940	7,290	8,910	10,530	12,690	15,120	18,360	22,140	27,000
3,3	0,330	0,396	0,495	0,594	0,726	0,891	1,089	1,287	1,551	1,848	2,244	2,706	3,300	3,960	4,950	5,940	7,260	8,910	10,890	12,870	15,510	18,480	22,440	27,060	33,000
3,9	0,390	0,468	0,585	0,702	0,858	1,053	1,287	1,521	1,833	2,184	2,652	3,198	3,900	4,680	5,850	7,020	8,580	10,530	12,870	15,210	18,330	21,840	26,520	31,980	39,000
4,7	0,470	0,564	0,705	0,846	1,034	1,269	1,551	1,833	2,209	2,632	3,196	3,854	4,700	5,640	7,050	8,460	10,340	12,690	15,510	18,330	22,090	26,320	31,960	38,540	47,000
5,6	0,560	0,672	0,840	1,008	1,232	1,512	1,848	2,184	2,632	3,136	3,808	4,592	5,600	6,720	8,400	10,080	12,320	15,120	18,480	21,840	26,320	31,360	38,080	45,920	56,000
6,8	0,680	0,816	1,020	1,224	1,496	1,836	2,244	2,652	3,196	3,808	4,624	5,576	6,800	8,160	10,200	12,240	14,960	18,360	22,440	26,520	31,960	38,080	46,240	55,760	68,000
8,2	0,820	0,984	1,230	1,476	1,804	2,214	2,706	3,198	3,854	4,592	5,576	6,724	8,200	9,840	12,300	14,760	18,040	22,140	27,060	31,980	38,540	45,920	55,760	67,240	82,000
10	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000	12,000	15,000	18,000	22,000	27,000	33,000	39,000	47,000	56,000	68,000	82,000	100,00

$$\hat{C}_1 = 3.3\text{F}, \hat{C}_2 = 6.8\text{F}, \hat{R}_1 = 0.22\Omega, \hat{R}_2 = 0.22\Omega$$



The normalized filter looks agreeable. Next, we need to scale the cut-off frequency to 1000 rad/s.

The frequency-normalized filter:

Multiplying in the numerator and denominator by a **frequency scaling factor** K_F^2 :

Substituting new variables:

$$\omega \stackrel{\text{def}}{=} K_F \hat{\omega} \quad C' \stackrel{\text{def}}{=} \frac{\hat{C}}{K_F} \quad R' \stackrel{\text{def}}{=} \hat{R}$$

We multiply the frequency by the same factor as we divide the capacitor. Hence the **impedance of the capacitor is unchanged** after frequency scaling.

$$H_{LP}(j\hat{\omega}) = \frac{\frac{K}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}{(j\hat{\omega})^2 + j\hat{\omega} \left(\frac{1}{\hat{R}_1 \hat{C}_2} + \frac{1}{\hat{R}_2 \hat{C}_2} + \frac{(1-K)}{\hat{R}_2 \hat{C}_1} \right) + \frac{1}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}$$

$$H_{LP}(j\hat{\omega}) = \frac{\frac{K \cdot K_F^2}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}{(jK_F \hat{\omega})^2 + jK_F \hat{\omega} \left(\frac{K_F}{\hat{R}_1 \hat{C}_2} + \frac{K_F}{\hat{R}_2 \hat{C}_2} + \frac{K_F(1-K)}{\hat{R}_2 \hat{C}_1} \right) + \frac{K_F^2}{\hat{R}_1 \hat{R}_2 \hat{C}_1 \hat{C}_2}}$$

$$H_{LP}(j\omega) = \frac{\frac{K}{R'_1 R'_2 C'_1 C'_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R'_1 C'_2} + \frac{1}{R'_2 C'_2} + \frac{(1-K)}{R'_2 C'_1} \right) + \frac{1}{R'_1 R'_2 C'_1 C'_2}}$$

$$\omega C' = \hat{\omega} \hat{C} \Rightarrow \frac{1}{j\omega C'} = \frac{1}{j\hat{\omega} \hat{C}} \Rightarrow \frac{1}{j\omega_c C'} = \frac{1}{j\hat{\omega}_c \hat{C}}$$

Now let us move the cut-off frequency to 1000 rad/s.

The frequency scaling needed:

$$K_F = \frac{\omega_c}{\hat{\omega}_c} = \frac{10^3 \text{ rad/s}}{1 \text{ rad/s}} = 1000$$

The frequency scaling of capacitors:

$$C' = \frac{\hat{C}}{K_F}$$

$$C'_1 = \frac{3.3 \text{ F}}{1000} = 3.3 \text{ mF}$$

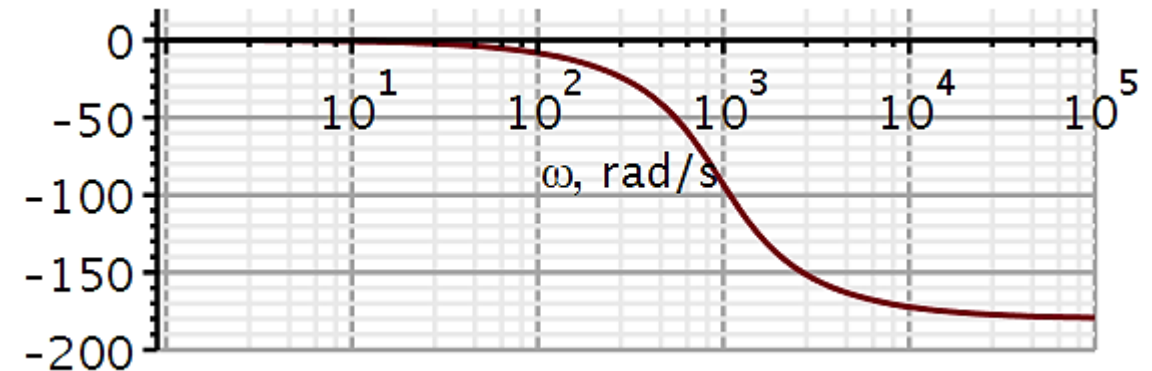
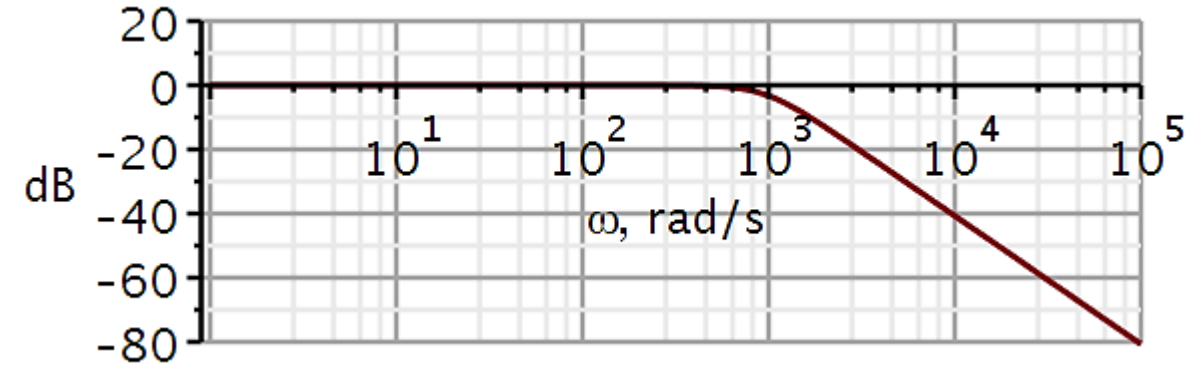
$$C'_2 = 6.8 \text{ mF}$$

Resistors are not frequency scaled.

$$R' = \hat{R}$$

Plotting the frequency characteristics after frequency scaling confirms that the cut-off frequency has moved to the desired value, but nothing else has changed.

Frequency scaling:



The capacitor values are far too big. We would like them to be in the nF range.

We are free to scale the components, as long as we do not change the time constants. Hence, we can multiply the resistors with a scale factor, provided we divide the capacitors by the same scale factor.

$$R \stackrel{\text{def}}{=} K_Z R' \quad C \stackrel{\text{def}}{=} \frac{C'}{K_Z}$$

$$\tau = RC = R'C' = \tau'$$

$$H_{LP}(j\omega) = \frac{\frac{K}{R'_1 R'_2 C'_1 C'_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R'_1 C'_2} + \frac{1}{R'_2 C'_2} + \frac{(1-K)}{R'_2 C'_1} \right) + \frac{1}{R'_1 R'_2 C'_1 C'_2}}$$

$$H_{LP}(j\omega) = \frac{\frac{K}{K_Z R'_1 K_Z R'_2 \frac{C'_1}{K_Z} \frac{C'_2}{K_Z}}}{(j\omega)^2 + j\omega \left(\frac{1}{K_Z R'_1 \frac{C'_2}{K_Z}} + \frac{1}{K_Z R'_2 \frac{C'_2}{K_Z}} + \frac{(1-K)}{K_Z R'_2 \frac{C'_1}{K_Z}} \right) + \frac{1}{K_Z R'_1 K_Z R'_2 \frac{C'_1}{K_Z} \frac{C'_2}{K_Z}}}$$

$$H_{LP}(j\omega) = \frac{\frac{K}{R_1 R_2 C_1 C_2}}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

Impedance scaling

We choose an impedance scaling which preserves the E6/E12 values.

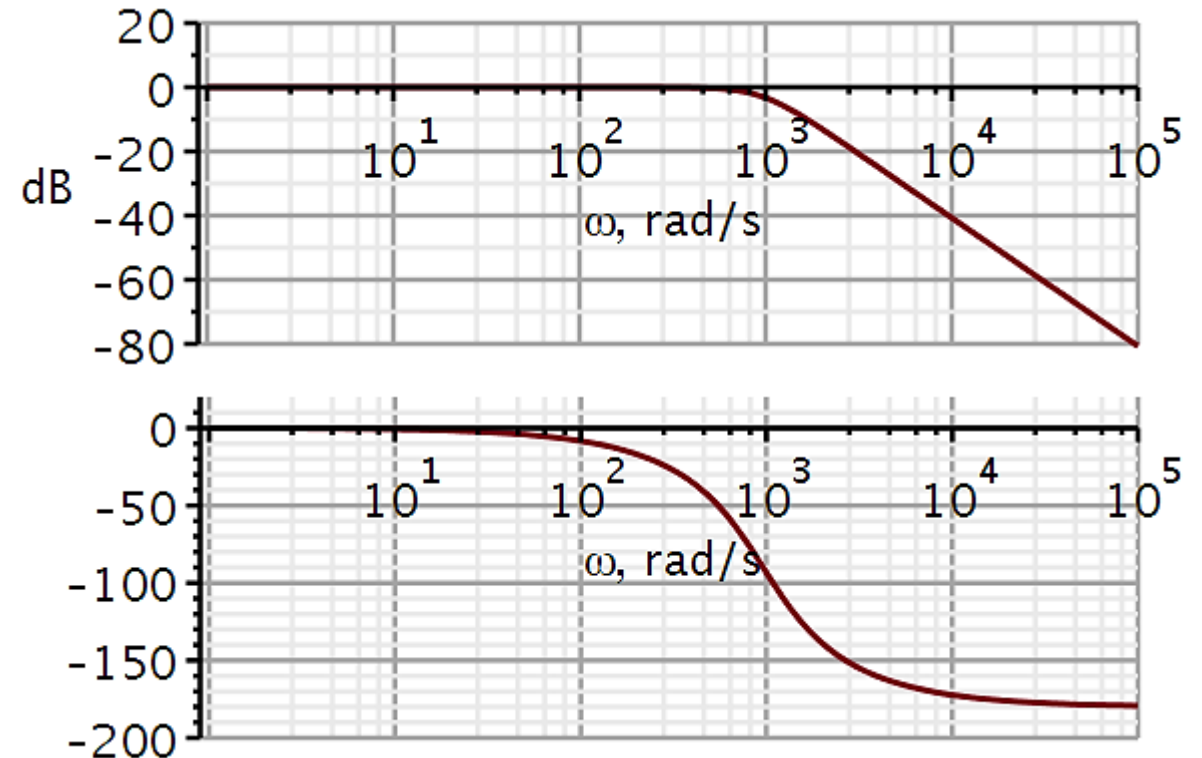
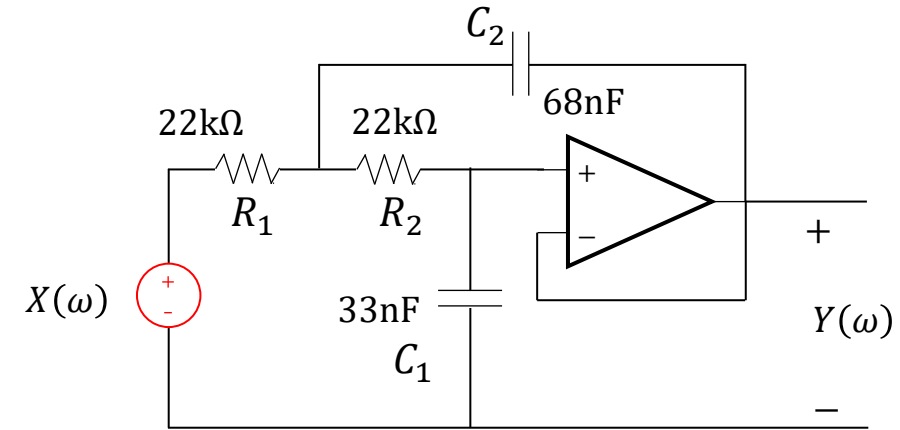
$$C_1 \stackrel{\text{def}}{=} 33\text{nF}$$

$$K_z = \frac{C'_1}{C_1} = \frac{3.3\text{mF}}{33\text{nF}} = 10^5$$

$$C_2 = \frac{C'_2}{K_z} = \frac{6.8\text{mF}}{10^5} = 68\text{nF}$$

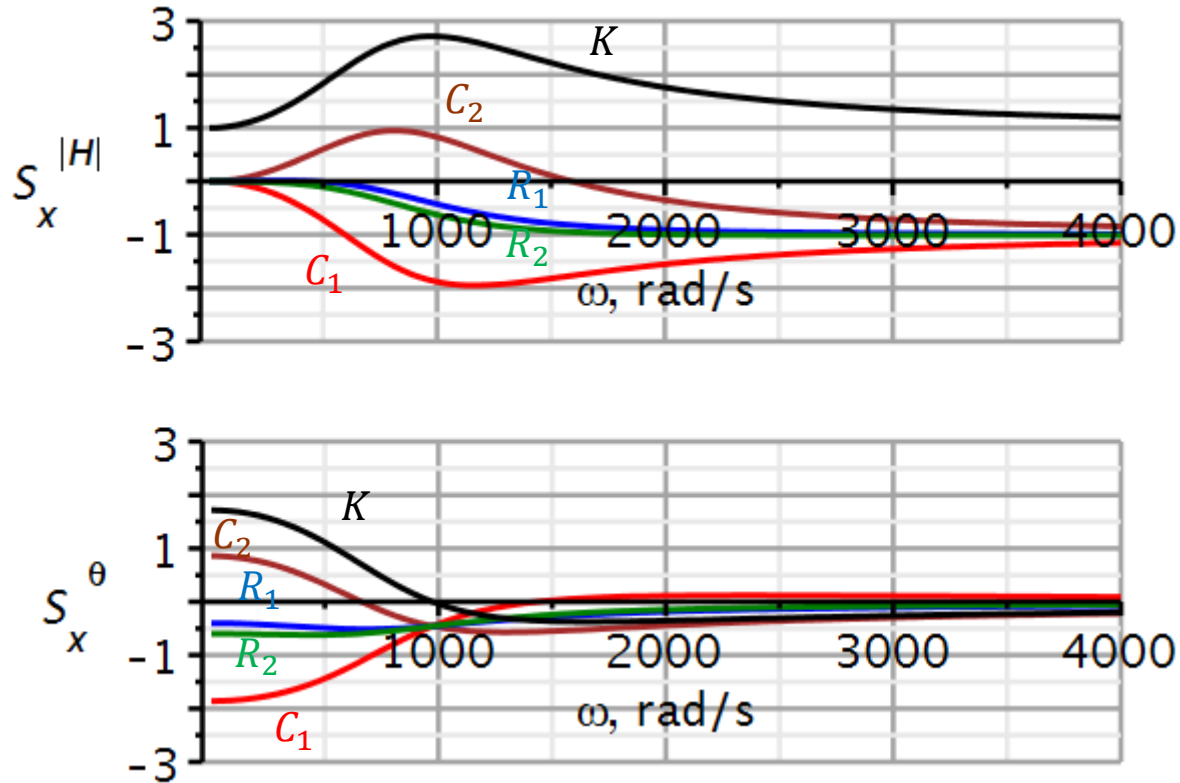
$$R_1 = K_z \cdot R'_1 = 10^5 \cdot 0.22\Omega = 22\text{k}\Omega$$

$$R_2 = K_z \cdot R'_2 = 10^5 \cdot 0.22\Omega = 22\text{k}\Omega$$

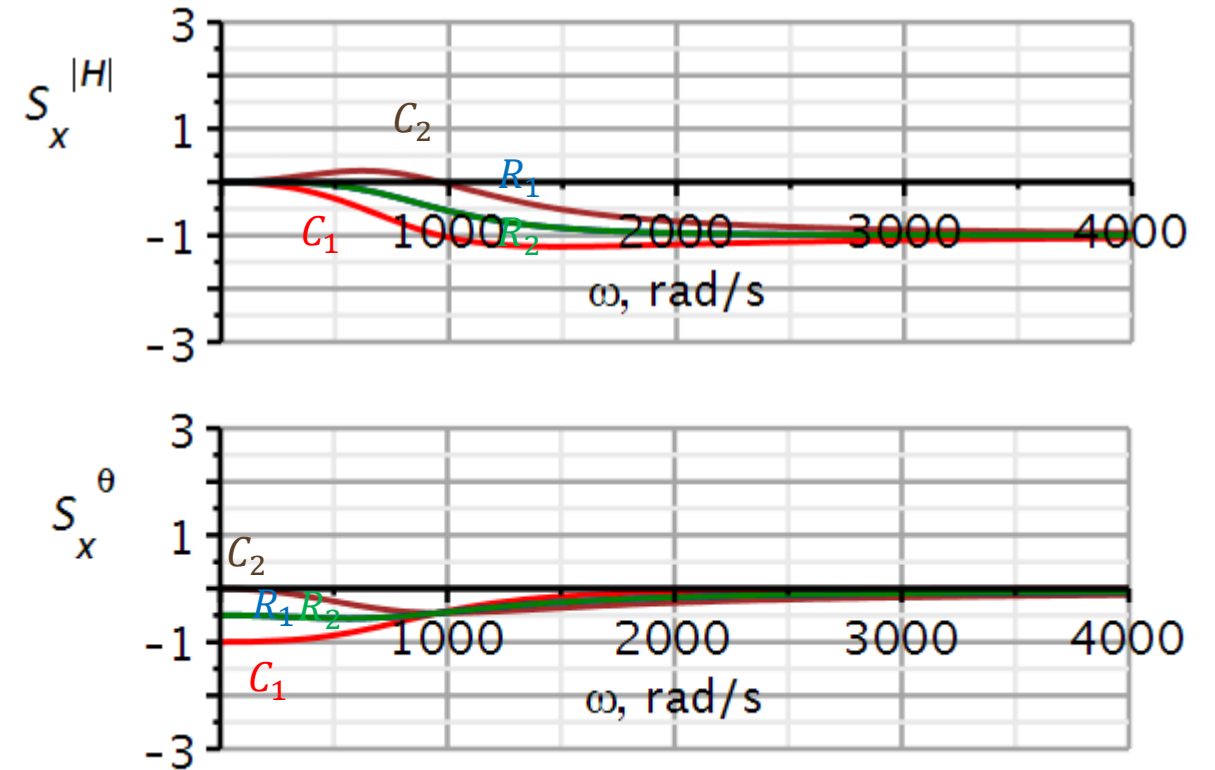


Design procedure for low sensitivity

Sensitivity for original design



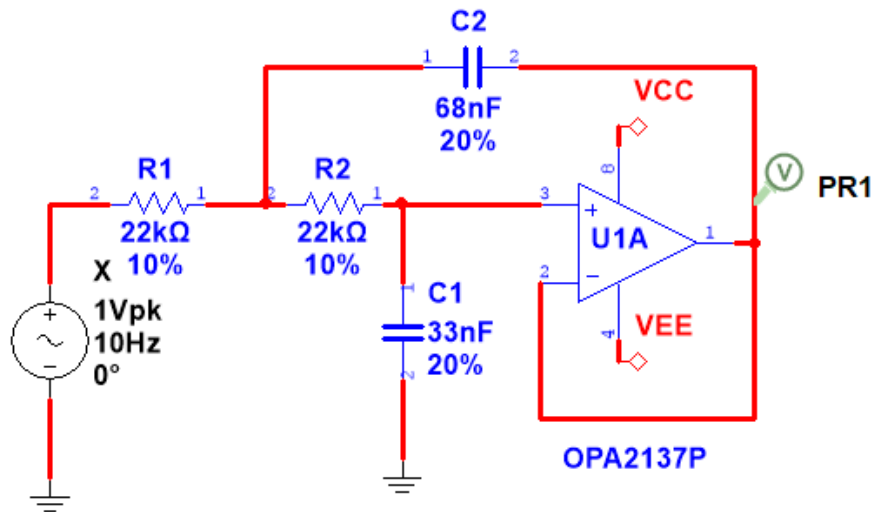
Sensitivity for improved design



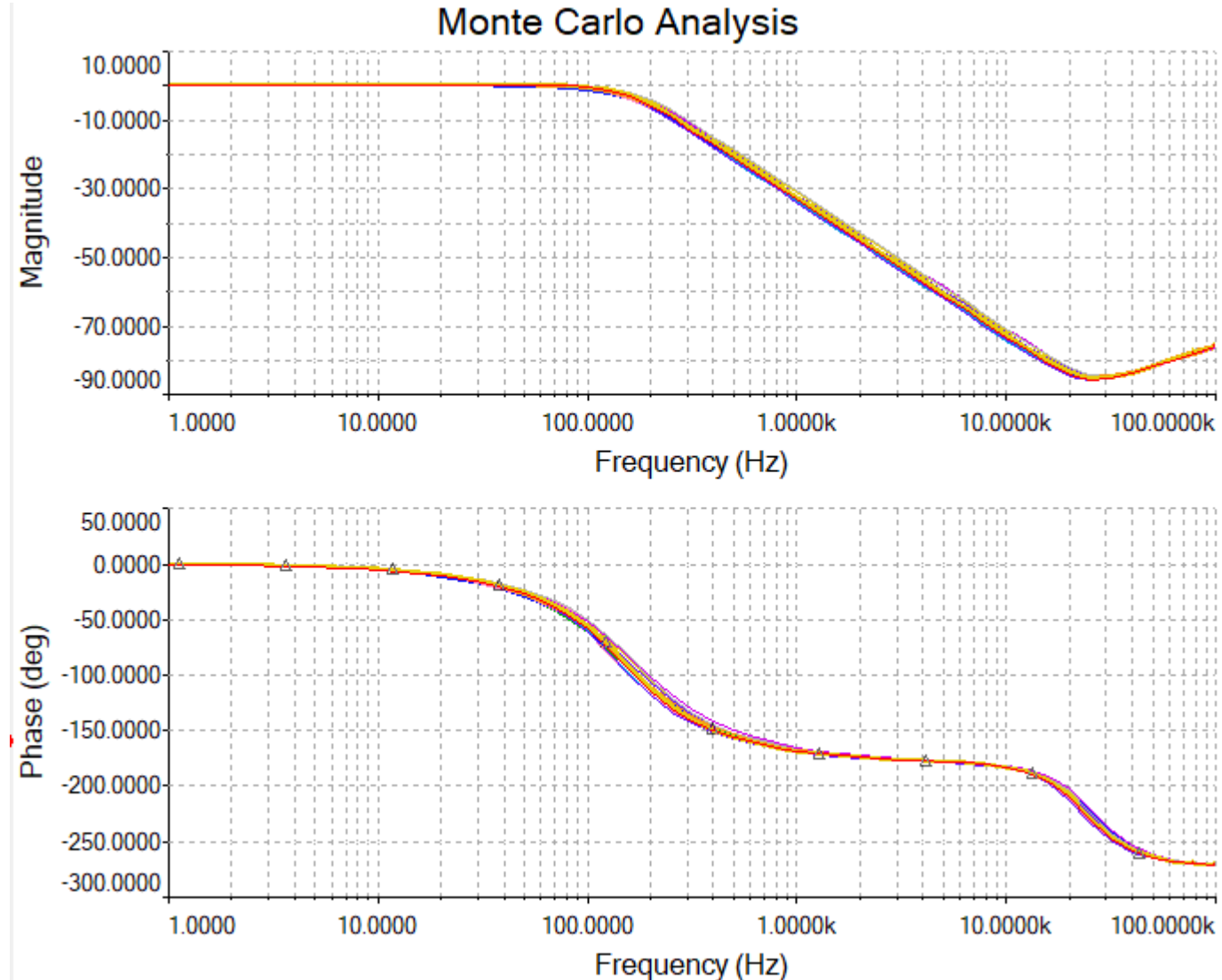
Our new design procedure has reduced sensitivities significantly.

Monte Carlo simulations are used in some circuit simulators to visualize the sensitivity to component change.

The simulator changes component values using a specified probability distribution and component range.

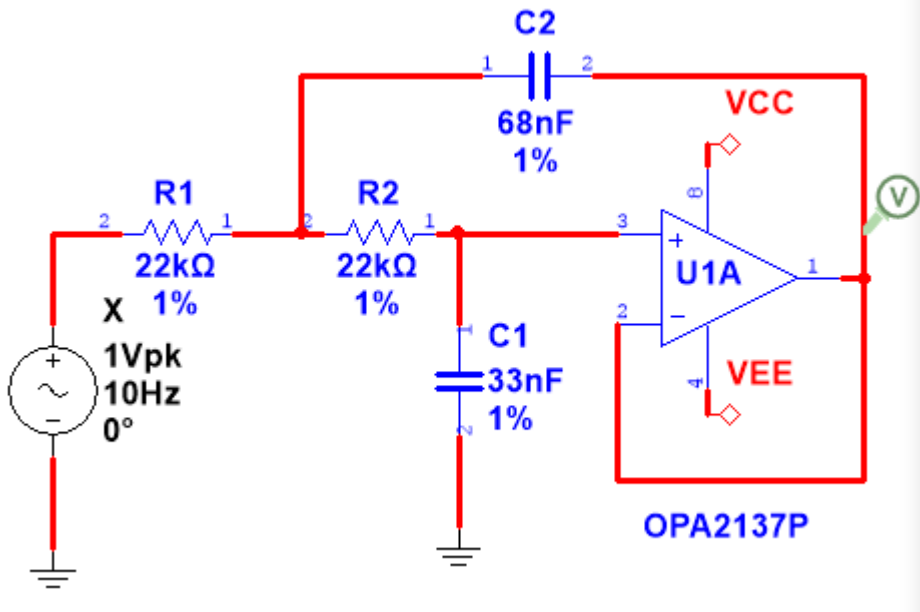


It gives the **most likely variation**, not the worst-case variation.

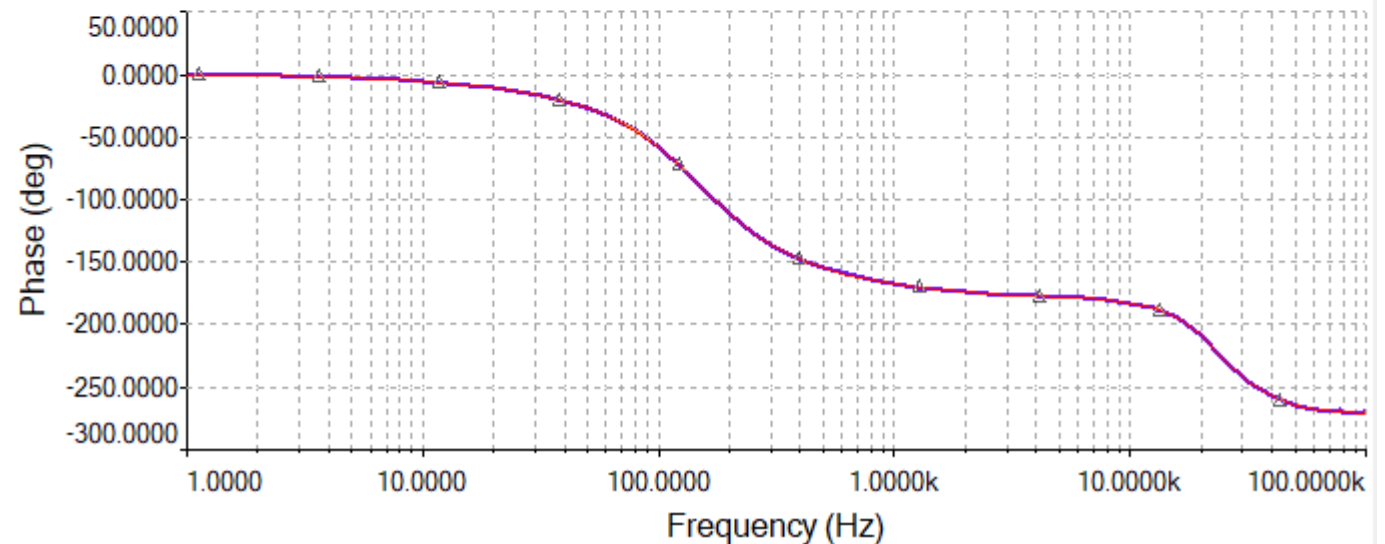
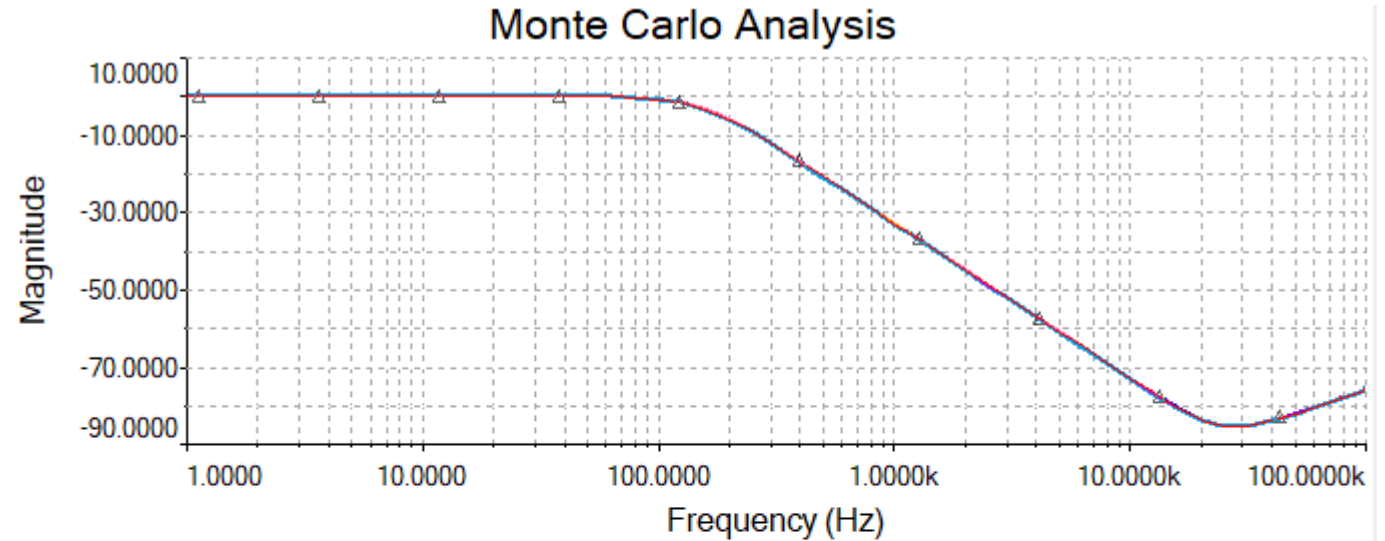


Lowering the tolerance on the components, lowers the variability.

By **lowering the tolerance** on components **one by one**, it is possible to single out the components which benefits the most from low tolerance.



Not available in KiCad

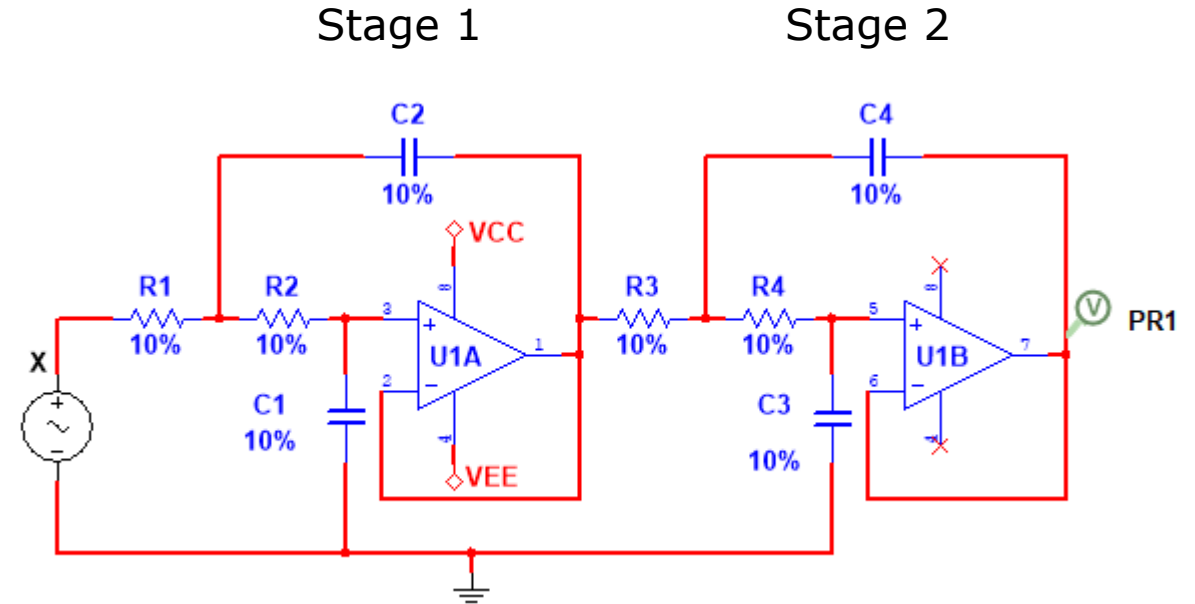


Problems

Problem 1 – EMG lowpass filter

Design a 4th order Butterworth lowpass filter with unity gain and a cut-off frequency of 500 Hz.

- Use two Sallen-Key stages in cascade. The filter should be very insensitive to component drift.
- Draw the amplitude and phase spectra in Maple/Python for:
 - the frequency-normalized filter
 - final design. Draw the spectra for each stage and for the combination.
- Use resistors from the E12 series and capacitors from the E6 series as much as possible, but not at all costs.

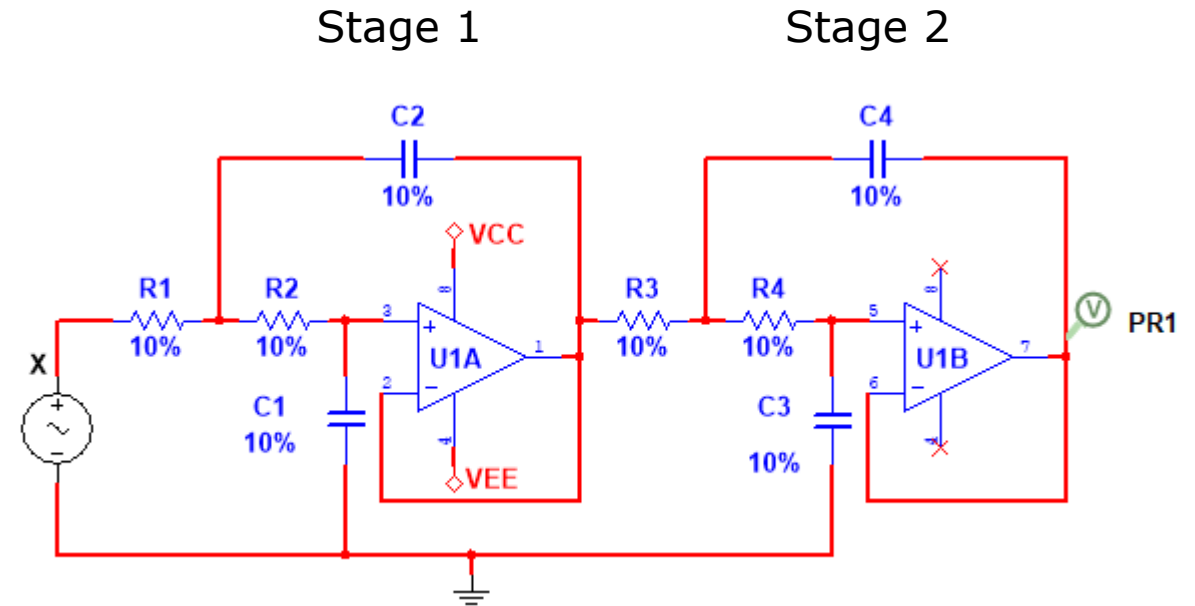


Solutions

Problem 1 – EMG lowpass filter

Design a 4th order Butterworth lowpass filter with unity gain and a cut-off frequency of 500 Hz.

- Use two Sallen-Key stages in cascade. The filter should be very insensitive to component drift.
- Draw the amplitude and phase spectra in Maple/Python for:
 - the frequency-normalized filter
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- Use resistors from the E12 series and capacitors from the E6 series as much as possible, but not at all costs.



Problem 1 – EMG lowpass filter (sol)

Stage 1:

Looking up in the table of Butterworth coefficients:

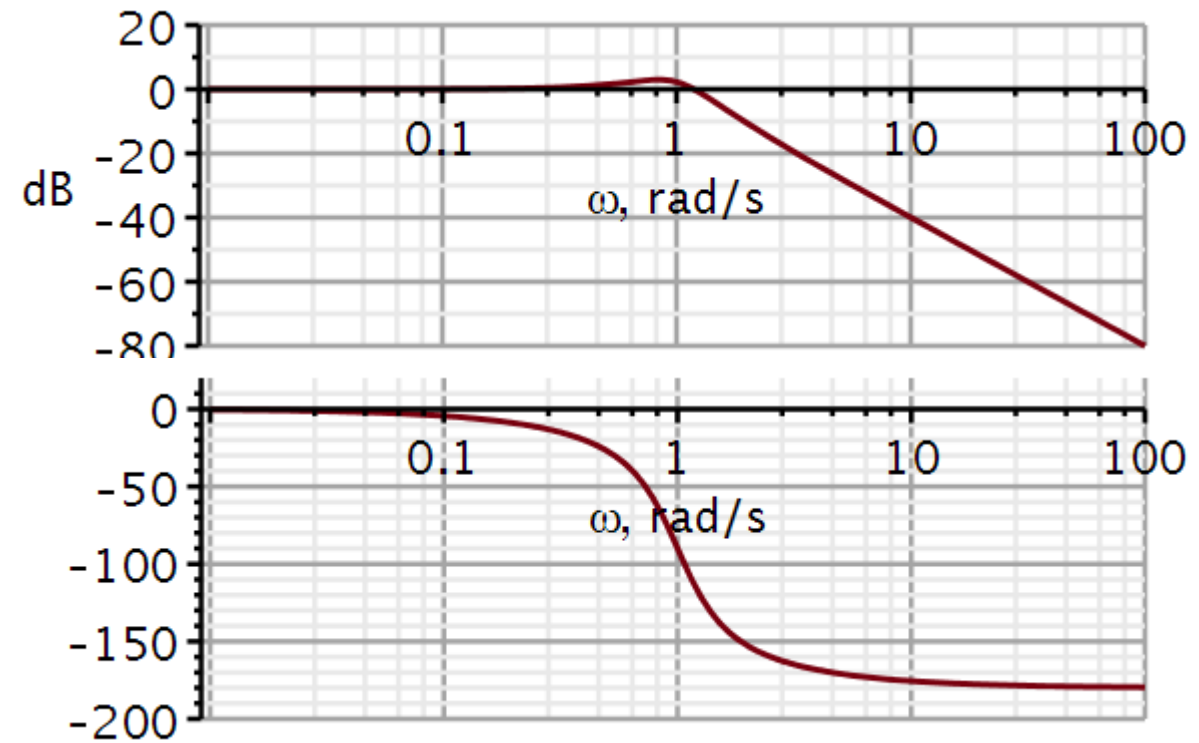
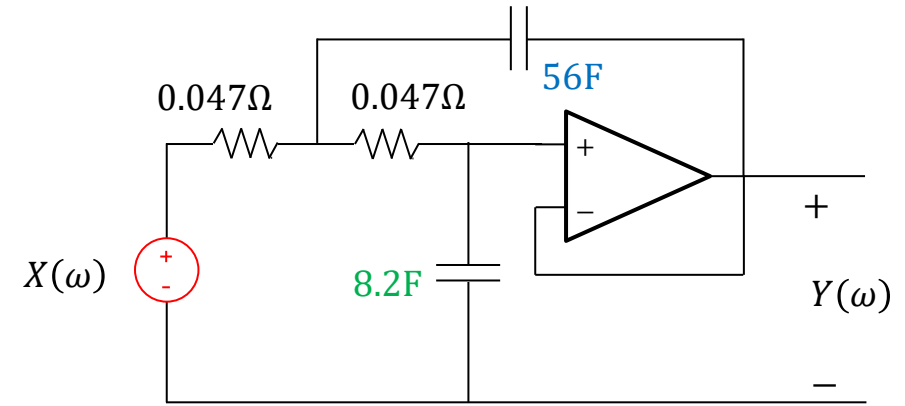
$$a_1 = 0.765367$$

$$a_0 = 1$$

Using the guidelines from sensitivity analysis:

$$\frac{\hat{C}_2}{\hat{C}_1} = \frac{4a_0}{a_1^2} = \frac{4}{(0.765367)^2} = 6.82842 \approx \frac{56\text{F}}{8.2\text{F}} = 6.82927$$

$$\hat{R}_1 = \hat{R}_2 = \frac{a_1}{2a_0\hat{C}_1} = 0.0466687\Omega$$



Problem 1 – EMG lowpass filter (sol)

Frequency scaling

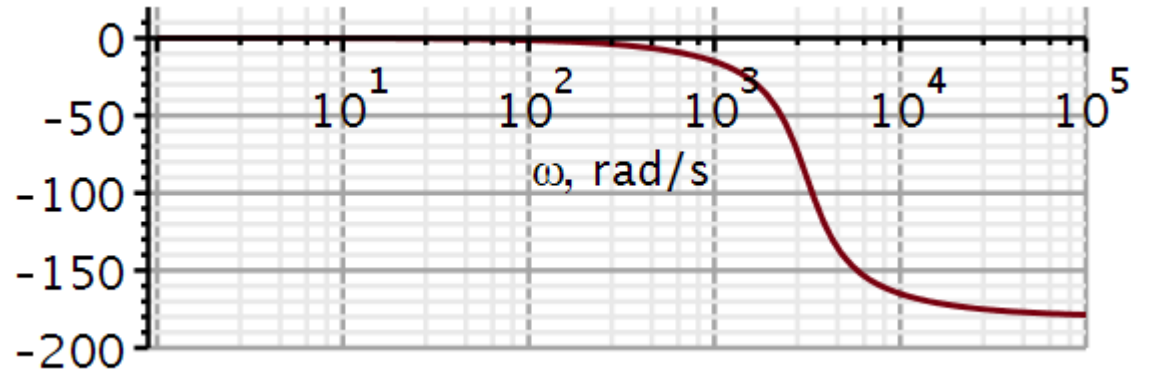
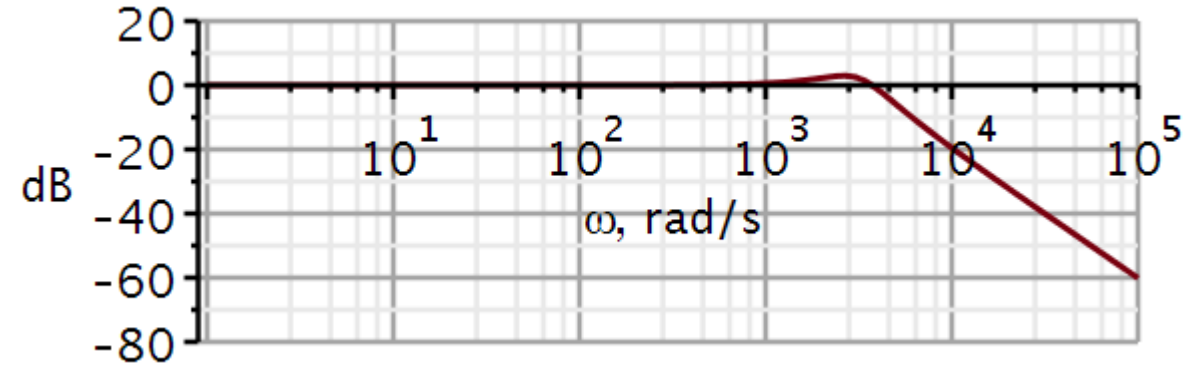
$$K_F = 2\pi \cdot 500 = 3141.592$$

$$C'_1 = \frac{\hat{C}_1}{K_F} = \frac{8.2\text{F}}{3141.592} = 2.610\text{mF}$$

$$C'_2 = \frac{\hat{C}_2}{K_F} = \frac{56\text{F}}{3141.592} = 17.83\text{mF}$$

$$R'_1 = R'_2 = \hat{R}_1 = \hat{R}_2 = 0.0466687\Omega$$

Stage 1



Problem 1 – EMG lowpass filter (sol)

In our first attempt at Impedance scaling, we seek to maintain the ratio of 56/8.2 for the capacitors.

Hence, we determine an impedance scaling factor, such that $C_1 = 8.2\text{nF}$ and $C_2 = 56\text{nF}$.

The resistors then scale to $15\text{k}\Omega$.

This works well, but 8.2nF is not in the E-6 series. On the next slide we seek an E-6 value.

$$C'_1 = 2.610\text{mF}$$

$$C'_2 = 17.83\text{mF}$$

$$R'_1 = R'_2 = 0.0466687\Omega$$

$$K_z = \frac{C'_1}{8.2 \cdot 10^{-9}\text{F}} = \frac{2.610\text{mF}}{8.2 \cdot 10^{-9}\text{F}} = 318292$$

$$C_2 = \frac{C'_2}{K_z} = \frac{17.83\text{mF}}{318292} = 56\text{nF}$$

$$R_1 = R_2 = 318292 \times 0.0466687\Omega = 14.85\text{k}\Omega \approx 15\text{k}\Omega$$

Problem 1 – EMG lowpass filter (sol)

Impedance scaling.

After frequency scaling, we have this product:

$$R'_1 C'_1 = 1.218 \cdot 10^{-4}$$

By looking up in the $R \cdot C$ look-up table earlier in the slide set, we can pick a base value for the resistor of 1.2 and a base value for the capacitor of 1.0 (in the E-6 series). Now we just need to find the correct magnitude for each.

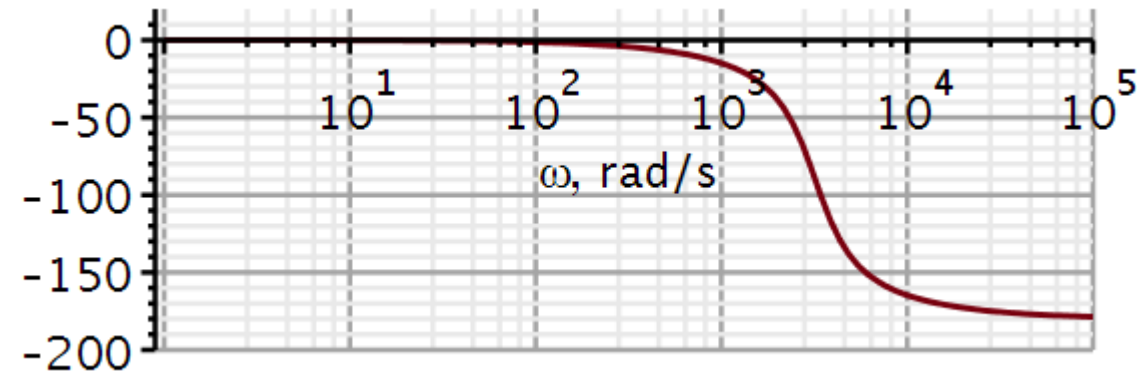
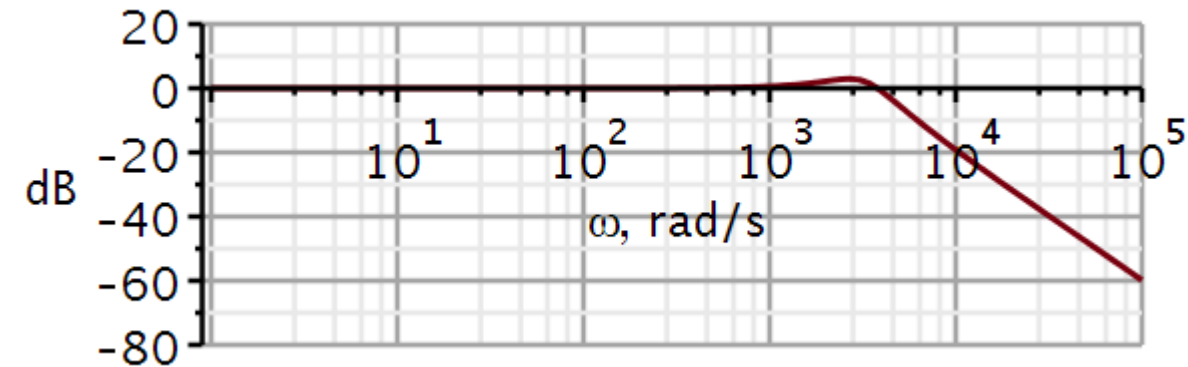
$$K_z = \frac{C'_1}{10 \cdot 10^{-9}\text{F}} = \frac{2.610\text{mF}}{10 \cdot 10^{-9}\text{F}} = 261014$$

$$C_1 = \frac{C'_1}{K_z} = 10\text{nF} \quad C_2 = \frac{C'_2}{K_z} = 68.29\text{nF} \approx 68\text{nF}$$

$$R_1 = R_2 = K_z R'_1 = 261014 \times 0.0466687\Omega = 12181\Omega \approx 12\text{k}\Omega$$

10nF and 68nF are both E-6 values, and 12k Ω is an E-12 value.

Stage 1



Problem 1 – EMG lowpass filter (sol)

Stage 2:

Looking up in the table of Butterworth coefficients:

$$a_1 = 1.84776$$

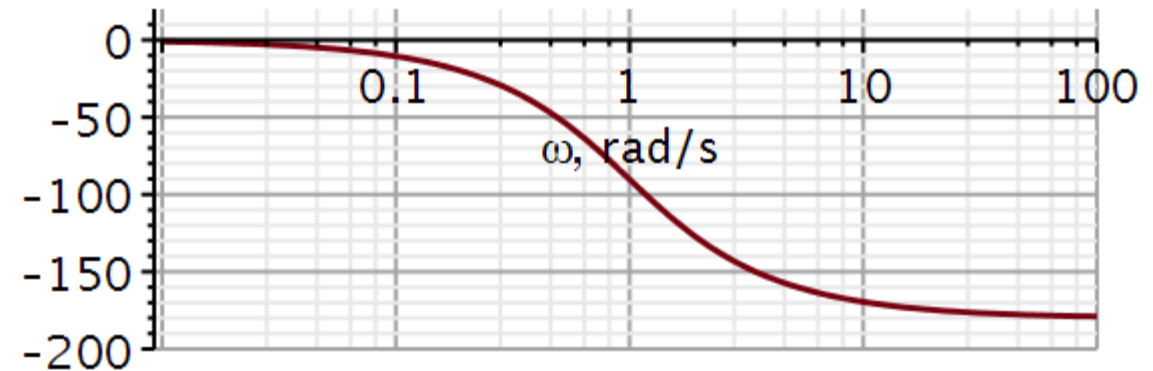
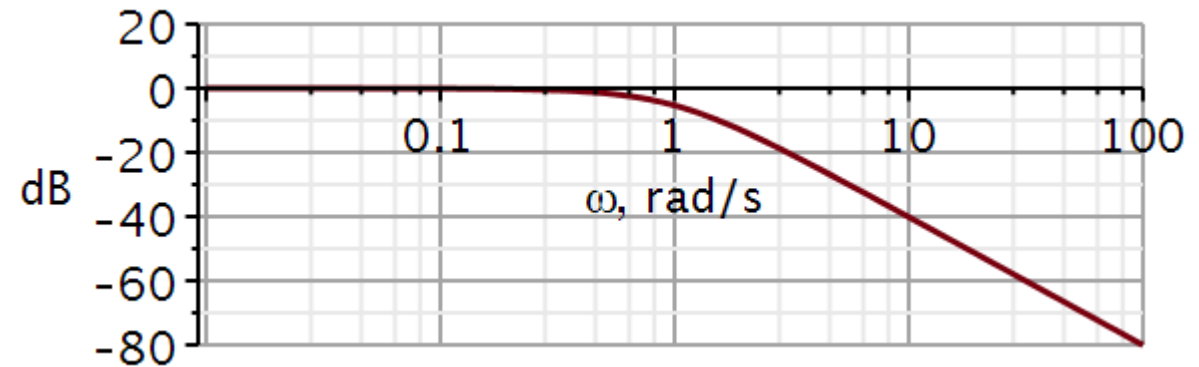
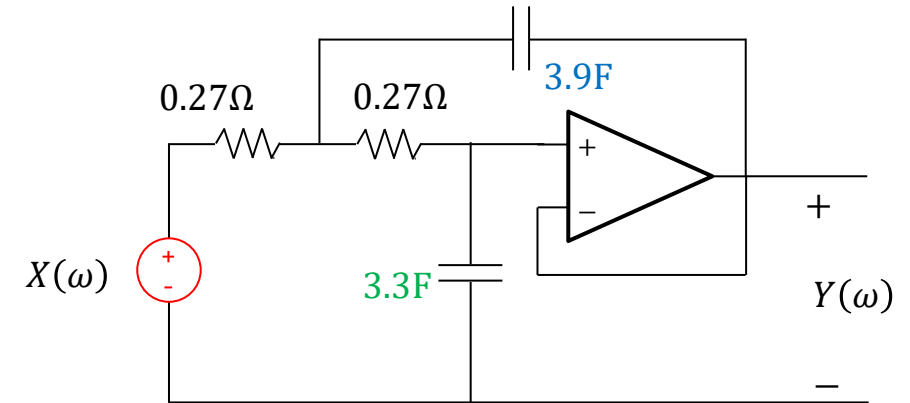
$$a_0 = 1$$

Using the guidelines from sensitivity analysis:

$$\frac{\hat{C}_4}{\hat{C}_3} = \frac{4a_0}{a_1^2} = \frac{4}{(1.84776)^2} = 1.17157 \approx \frac{3.9\text{F}}{3.3\text{F}} = 1.18182$$

$$\hat{R}_3 = \hat{R}_4 = \frac{a_1}{2a_0\hat{C}_3} = 0.27996\Omega$$

Stage 2



Problem 1 – EMG lowpass filter (sol)

Frequency scaling

$$\omega_n = \frac{1}{\sqrt{\hat{R}_3 \hat{R}_4 \hat{C}_3 \hat{C}_4}} = 0.99566$$

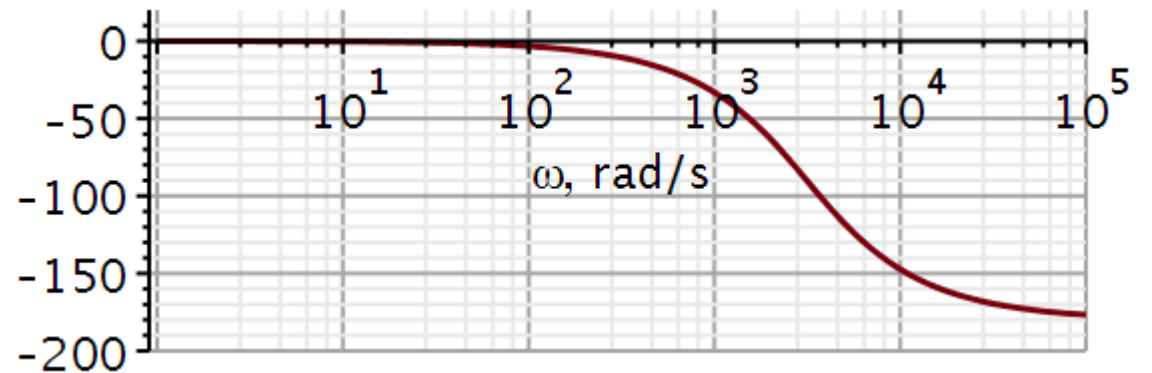
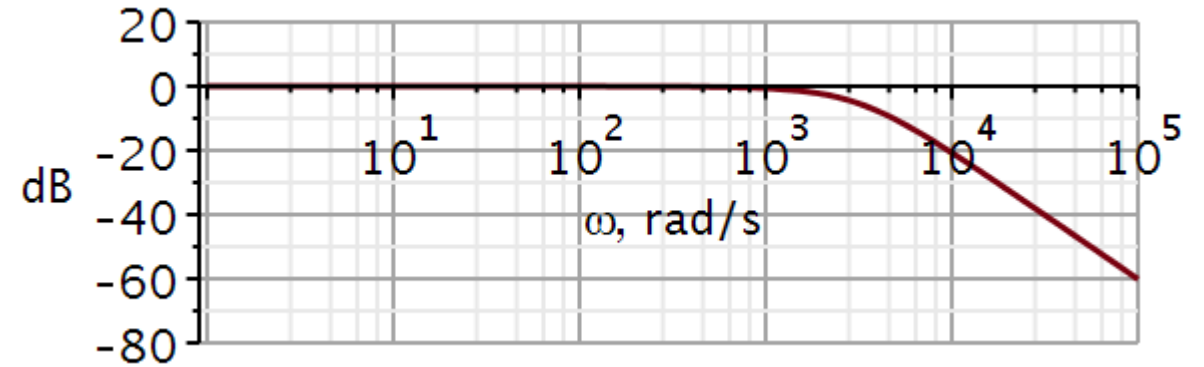
$$K_F = \frac{2\pi \cdot 500}{\omega_n} = 3155.3$$

$$C'_3 = \frac{\hat{C}_3}{K_F} = \frac{3.3\text{F}}{3155.3} = 1.046\text{mF}$$

$$C'_4 = \frac{\hat{C}_4}{K_F} = \frac{3.9\text{F}}{3155.3} = 1.236\text{mF}$$

$$R'_3 = R'_4 = \hat{R}_3 = \hat{R}_4 = 0.27996\Omega$$

Stage 2



Problem 1 – EMG lowpass filter (sol)

We have:

$$C'_3 = 1.046\text{mF}$$

$$C'_4 = 1.236\text{mF}$$

$$R'_3 = R'_4 = 0.27996\Omega$$

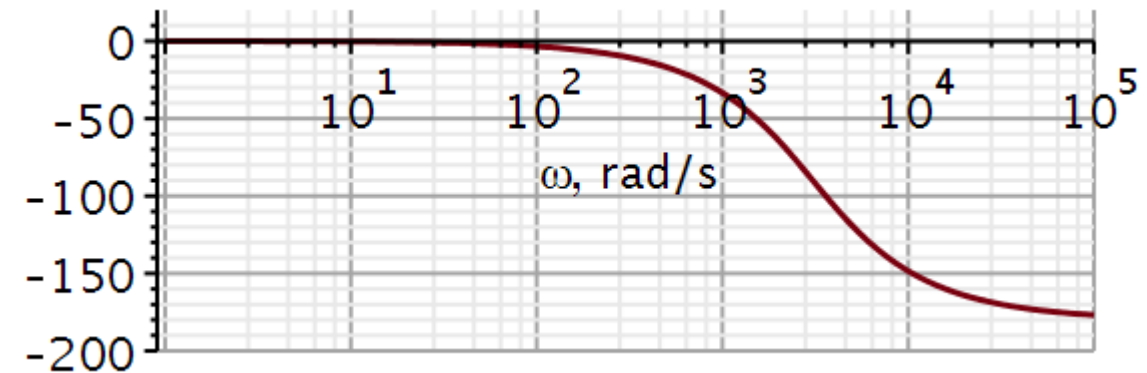
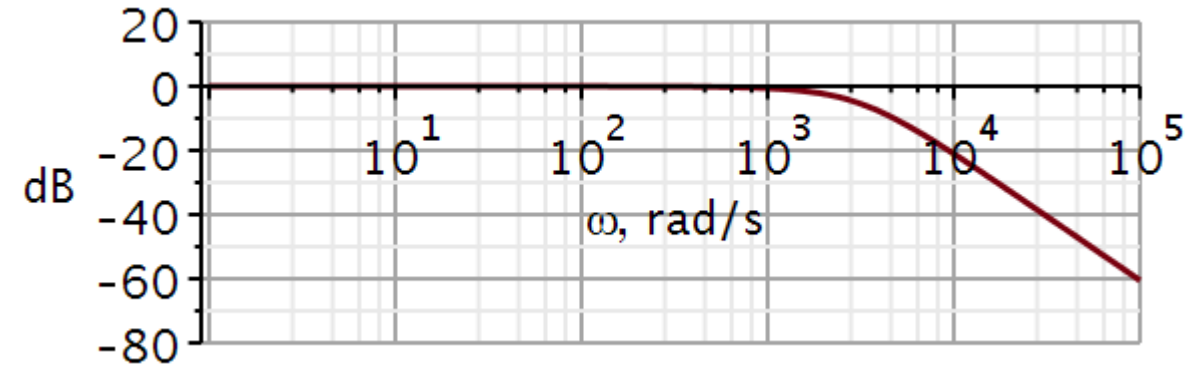
1.2 is not an E-6 value. Because all three values are close to E-12 values, we will use values from E-12.

$$K_z = \frac{C'_3}{10\text{nF}} = \frac{1.046\text{mF}}{10\text{nF}} = 104600.$$

$$C_4 = \frac{C'_4}{K_z} = \frac{1.236\text{mF}}{104600} = 11.8\text{nF} \approx 12\text{nF}$$

$$R_3 = R_4 = 104600 \times 0.27996\Omega = 29.2\text{k}\Omega \approx 27\text{k}\Omega$$

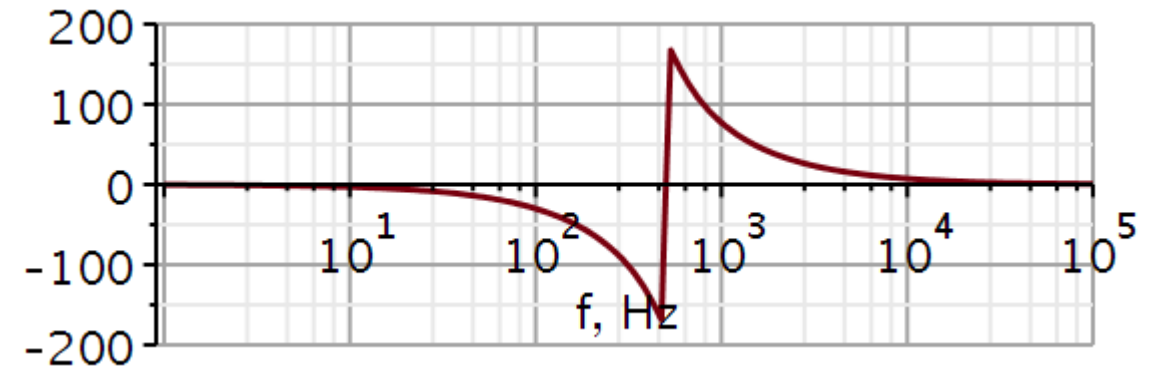
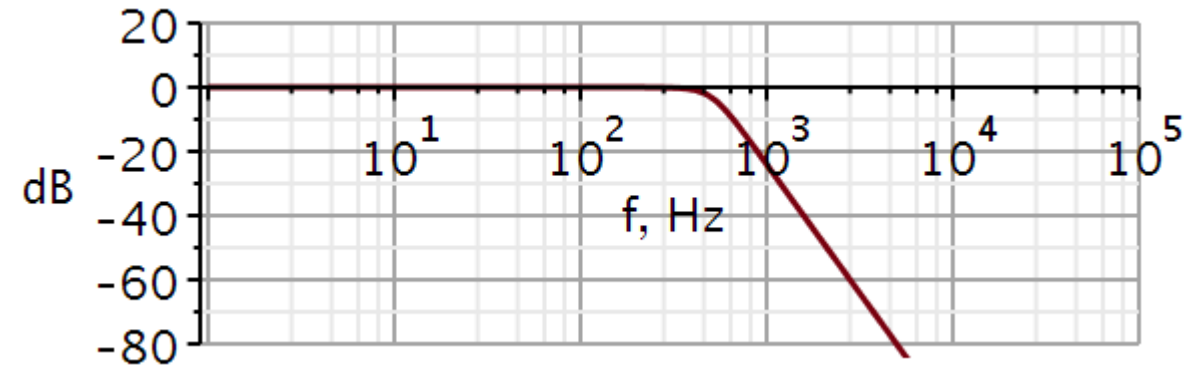
Stage 2



Problem 1 – EMG lowpass filter (sol)

$$H(\omega) = H_1(\omega) \cdot H_2(\omega)$$

In the final plots the frequency axis have been converted to Hz.



Problem 1 – EMG lowpass filter (sol)

We would like to compare the magnitude of the two individual stages with the combined system.

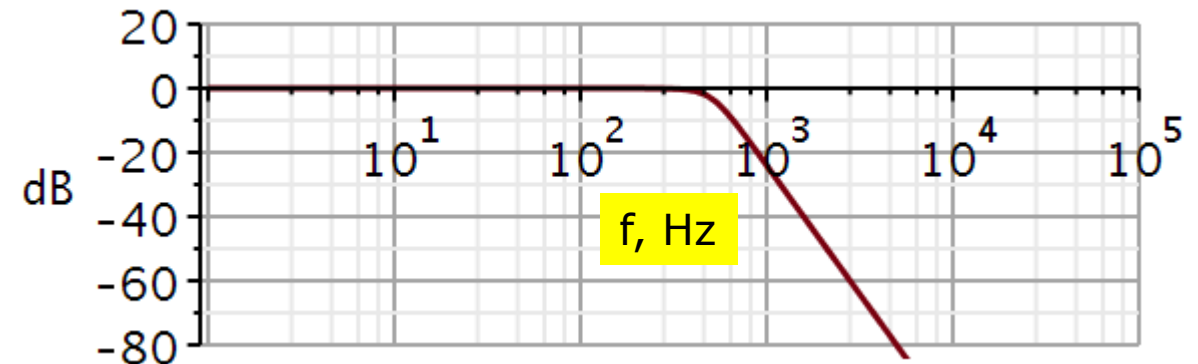
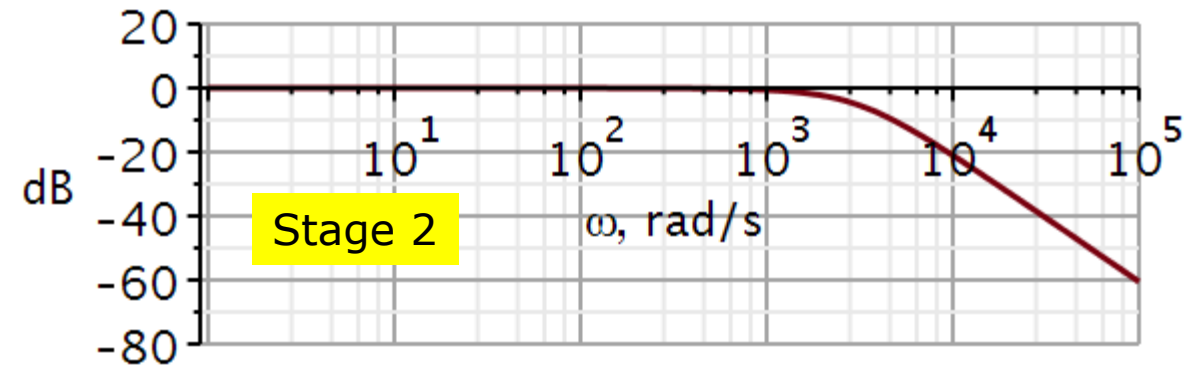
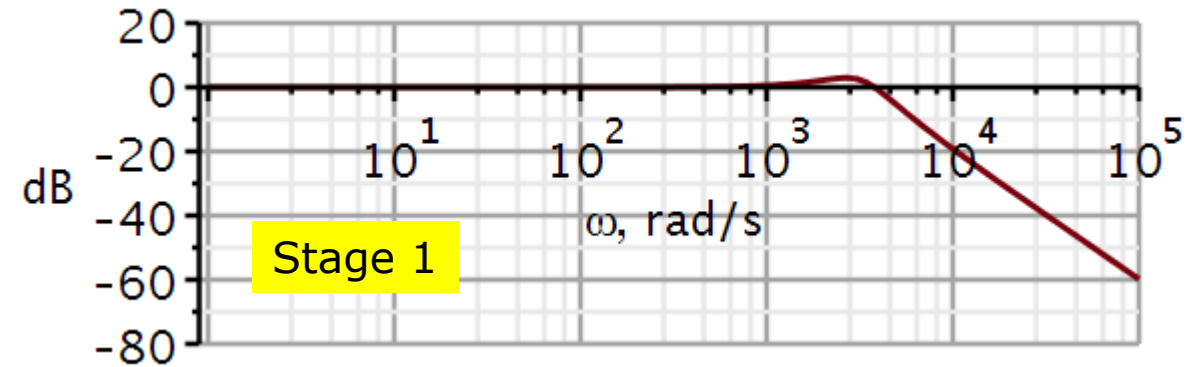
We observe a small peak in the magnitude of stage 1.

$$\begin{aligned} a_1 &= 0.765367 \\ a_0 &= 1 \end{aligned} \quad \zeta_1 = \frac{a_1}{2\sqrt{a_0}} = \frac{0.765}{2} = 0.38$$

For stage 2:

$$\begin{aligned} a_1 &= 1.84776 \\ a_0 &= 1 \end{aligned} \quad \zeta_2 = \frac{a_1}{2\sqrt{a_0}} = \frac{1.848}{2} = 0.92$$

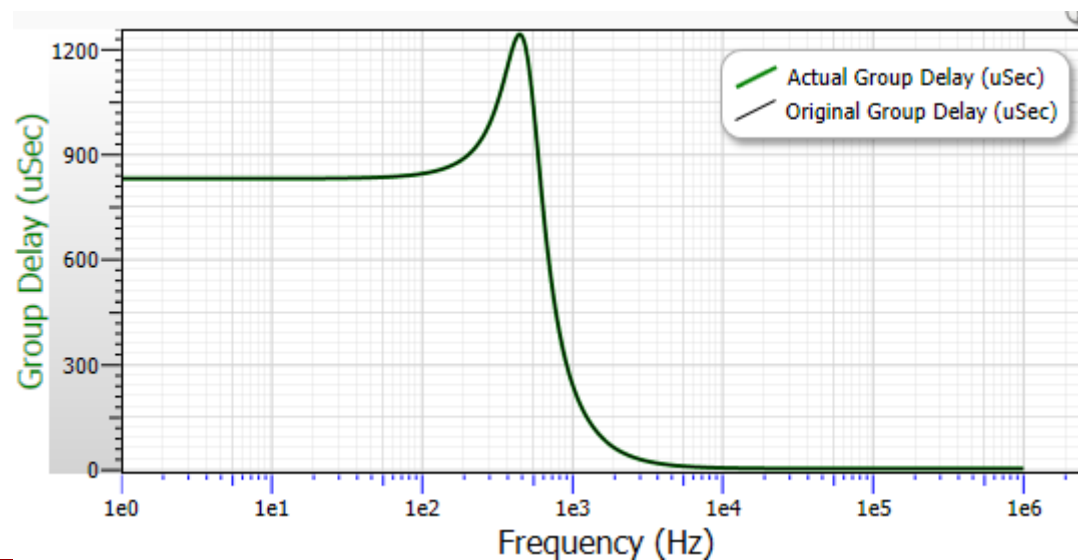
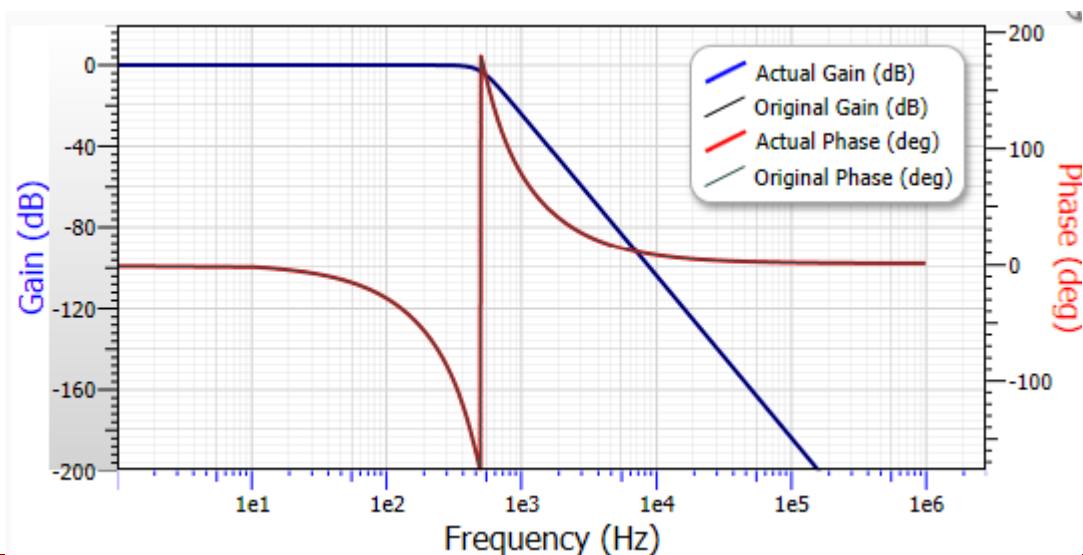
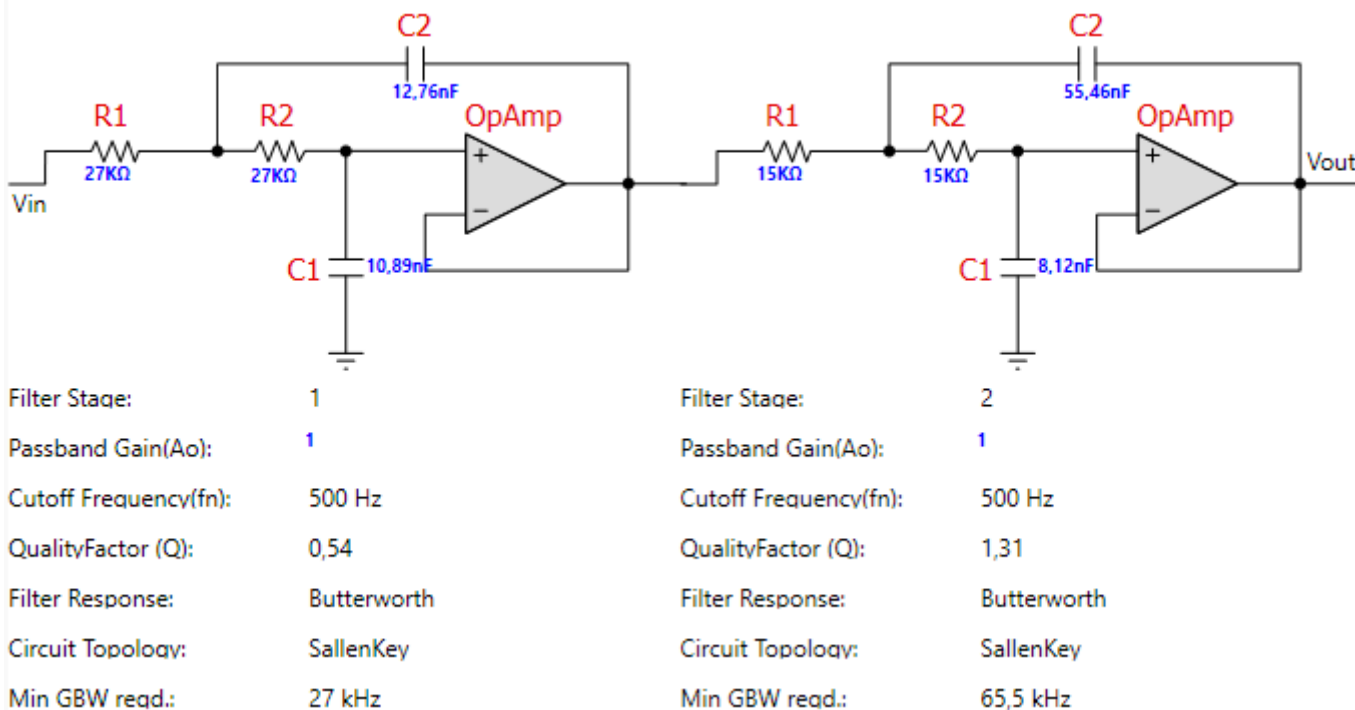
As expected of a 4th order filter. One pole pair lies below and the other pair above the 45° degree line in the complex plane. **Which is which?**



Problem 1 – EMG lowpass filter (sol)

Circuit components calculated in FilterPro (Texas Instruments).

Here the resistors in FilterPro have been changed to my own values. The capacitor values should then also change to match my values. And they do. They have not yet been rounded to E12 values.



Problem 1 – EMG lowpass filter (sol)

Monte Carlo analysis in Multisim using 10% tolerance on all components.

